Tecto

a Digital Fish from the Silicon Cambrian

A 128-Neuron Recurrent Network Recapitulates the Optic Tectum

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Abstract

A GRU gated recurrent network with 128 hidden units, trained atop a frozen, self-supervised visual encoder, achieves simultaneous foraging and obstacle avoidance in three dimensions using only binocular vision—without semantic labels, depth maps, or any coordinate information. We call it **Tecto**, a 128-neuron digital fish from the Silicon Cambrian. Within its training distribution, it reaches a 99.7% food-capture rate with a 1.3% obstacle-collision rate, and tracks randomly-walking food with an 82.8% success rate despite never being trained on moving targets. Systematic circuit analysis reveals that the network spontaneously converges to the same functional organization as the vertebrate optic tectum: the update gate z encodes behavioral intensity (neuromodulatory gain control), the reset gate r is structurally silent, foraging and avoidance are encoded along nearly orthogonal hidden-state directions, and the output matrices implement a distributed population-vector decoder. Ablating the top-16 hidden dimensions drops the food-capture success rate (ATE) from 98.0% to 80.7%; ablating the bottom-16 has zero effect, confirming ~12.5% of parameters as silent synapses. Fifteen generations of evolution fail to improve the pretrained brain—in a 594,468-dimensional parameter space the uphill probability of random mutation is below 10^{-5000}.

Reverse validation through targeted lesioning provides independent evidence for the model's biological fidelity. We systematically damaged ten circuit loci spanning sensory, gain-control, recurrent, and motor-output pathways. Each lesion produced a behavioral phenotype matching the clinical presentation of the corresponding biological damage: $\mathbf{b}_z$ shift generates a continuous bradykinesia-to-hyperkinesia spectrum with preserved task competence; unilateral $\mathbf{W}_2$ row ablation recapitulates contralateral saccadic deficits; sensory-column perturbation at δ/θ frequencies produces direction-specific resonance; and coherent noise in only 5% of neurons is sufficient to collapse population-code averaging. Every lesion maps to a known computational function in the characterized circuit.

We propose that a set of physical constraints—binocular vision, continuous time, finite neuron budget, real-time motor output—defines a low-dimensional feasible solution space whose stable solution is the functional organization of the optic tectum. This hypothesis makes a specific, falsifiable prediction: if the same constraints are imposed on a different encoder (e.g., CLIP), a different recurrent architecture (e.g., LSTM), or different motor dynamics, the same structural convergence should re-emerge. Should it fail to do so, the hypothesis is falsified. These substitution experiments have not yet been performed; the hypothesis currently rests on a single data point.

1 Introduction

1.1 Two Roads Diverged

Over the past three decades of embodied AI, the problem of visually-guided autonomous movement has been pursued along two divergent roads.

The **semantic road** builds explicit intermediate representations: object detectors, semantic segmentation maps, depth estimators, path planners. Each module requires human-annotated data and human-designed interfaces. These systems work brilliantly when conditions match exactly—which is why they dominate autonomous driving. But they are brittle, expensive, and, in a certain sense, not how biology operates.

The **end-to-end road** attempts to learn perception-to-action mappings directly, typically via reinforcement learning (RL) or behavioral cloning. In principle these systems collapse perception and control into a single learned function. In practice, however, they remain far from direct: every "end-to-end" pipeline inherits a collection of latent intermediate representations and oracle parameters—BEV grids, trajectory proposals, occupancy maps, hand- designed reward functions—that constrain the solution before learning begins. RL demands astronomical numbers of trial-and-error episodes to discover basic behaviors [1, 2, 3]. Behavioral cloning imports the fragility of causal misidentification [4], locking the policy inside the demonstrator's distribution.

Biology took a **third road**. The first vertebrates with eyes did not have ImageNet. They did not have backpropagation. They did not have 600 epochs of Q-learning. What they had was the superior colliculus—an ancient structure in the midbrain that receives input directly from the retina and projects directly to brainstem motor nuclei [5, 6, 7]. Nowhere in between is the proposition "I know this is food." There is only a continuous, distributed, population-coded sensory-to-motor mapping, honed by a half-billion years of selection in a world of carbon and water.

1.2 We Asked a Simple Question

Can we build such a system using a frozen, general-purpose visual encoder and a minimal trainable recurrent controller? And if we do, will that controller spontaneously develop a functional organization that resembles the one evolution converged on over a half-billion years of trial and error?

We answered this question by designing **Tecto**—a 128-neuron recurrent controller from the Silicon Cambrian. Tecto is the *simplest possible system* that satisfies the same physical constraints as the vertebrate tectum: binocular visual input, continuous-time dynamics, a finite neuron budget, and real-time motor output. It is not a neuroscience-inspired model; it is a minimal instantiation of the constraints. Whatever structure emerges is the product of training, not human design—we provided the loss function, backpropagation through time optimized the weights.

What we found is structural.

1.3 Related Work

The present work sits at the intersection of four research communities.

1.3.1 Frozen Visual Encoders for Robot Policies

A growing body of work demonstrates that self-supervised visual representations transfer effectively to robotic tasks when kept frozen. DINO-WM [8] builds world models directly on frozen DINOv2 patch features for zero-shot planning. DexGraspVLA [9] uses two frozen DINOv2 models for dexterous grasping, finding that frozen features outperform fine-tuned ones under domain shift. R3M [10] and VC-1 [11] offer general-purpose visual representations for manipulation. Several groups have independently confirmed that frozen DINOv2 outperforms end-to-end trained encoders under visual distribution shift [12, 13]; the WILDS benchmark suite [14] provides standardized evaluation protocols for such distribution-shift robustness. In model-based RL, partially-frozen DINOv2 features achieve the strongest out-of-domain generalization. **Gap:** none of these works use frozen encoders for *continuous, real-time* visuomotor control—they target discrete manipulation or navigation tasks.

1.3.2 End-to-End Visuomotor Control

The dominant paradigm combines RL with world models [2, 15] or behavioral cloning with diffusion policies [16]. CoPhy [17] addresses the imitation learning ceiling by distilling VLM knowledge into a BEV world model. At the extreme of efficiency, Tiny-PULP-Dronets [18] achieve autonomous drone navigation with 50× fewer parameters. RT-2 [19] has pushed vision-language-action models to new heights. In autonomous driving, end-to-end approaches [20] and planning-oriented methods [21, 22] have demonstrated that direct perception-to-action pipelines can outperform modular stacks. **Gap:** despite the label "end-to-end," these systems operate on engineered intermediate representations (BEV features, trajectory anchors, occupancy grids) rather than raw sensory input. All require either massive environment interaction budgets (RL) or curated human demonstrations (imitation). None train a continuous visuomotor policy on static oracle supervision with zero environment interaction—the regime in which Tecto operates.

1.3.3 Computational Models of the Optic Tectum

The zebrafish visual system provides the most complete model organism for tectal circuit analysis [23], with transparent larval stages enabling whole-brain calcium imaging of retinotectal processing. Recent bio-physically constrained reservoir models built from the zebrafish tectum connectome [24] demonstrate that different tectal interneuron subtypes control sensorimotor accuracy and robustness. A 2024 study [25] showed that collicular motor units respond to kinetic visual features rather than static spatial receptive fields. Spiking neural network models [26] reproduce saccade kinematics using adaptive exponential integrate-and-fire neurons. Earlier theoretical work on neural population geometry [27, 28] and probabilistic population codes [29, 30] provides the mathematical framework for distributed sensorimotor representations. **Gap:** existing tectum models are either bio-physically constrained reservoirs that embed connectomic detail without exposing internal circuit logic, or abstract mathematical frameworks that describe population coding principles without instantiating them in a working visuomotor controller. None has been probed through the four-phase circuit analysis performed here—gate-level mechanistic decomposition, hidden-state geometry, weight-pathway tracing, and causal ablation—which is what reveals the spontaneous emergence of tectal organization from training alone.

1.3.4 Circuit Interpretability of Recurrent Networks

The mechanistic interpretability community has recently expanded to recurrent architectures. A NeurIPS 2025 study [31] introduced time-resolved circuit discovery via windowed causal interventions, revealing excitatory-dominated memory circuits and inhibitory-gated state transitions. Systematic decomposition of gating, attention, and recurrence [32] found that neural gating plus attention improves all standard RNNs. Optimal ablation methodology [33] establishes principled baselines for component removal. Population-coded spiking networks [34] extend distributed encoding to energy-efficient robotic control. **Gap:** no prior work has performed systematic circuit analysis on a visuomotor recurrent network—probing gates across behavioral scenarios, analyzing hidden-state geometry, and causally testing functional specialization through targeted ablation.

1.4 Core Contributions

Five claims emerge from the experiments reported below:

1. **Behavioral generalization without dynamic training.** A GRU(128) trained on static food-obstacle data from a heuristic controller—never touching a moving target—achieves 99.7% ATE with 1.3% collision rate in mixed environments (300 fish, 400 steps), a qualified ATE of 82.8% when tested against randomly-walking food (500 fish), and smooth tracking degradation under a food-speed sweep. Dynamic tracking *emerges* from static training—the GRU does not "switch modes"; it smoothly interpolates its population code.

2. **The GRU's organization spontaneously converges to the optic tectum.** Systematic circuit analysis—gate probing, hidden-state manifold, weight-pathway tracing, and ablation of all 128 hidden dimensions—reveals: (a) food-seeking and obstacle-avoidance occupy nearly orthogonal hidden-state directions ($\cos \approx -0.047$), (b) output decoding is distributed—"turn" and "forward" have no dedicated neurons, (c) 197K of the 594K trainable parameters (33%) are unused: the reset gate r is inactive under all conditions (constant 0.500, $\sigma = 0.000$; ~197K parameters), and 16/128 hidden dimensions carry zero effective output weight (ablation produces no effect; ~0.5K parameters).

3. **Supervised efficiency exceeds evolution by an astronomically wide margin.** Fifteen generations of natural selection starting from the pretrained brain produce no systematic improvement in a modern architecture (population mean fitness shows no trend after 15 generations). The reason is rooted in high-dimensional geometry: in a 594,468-dimensional parameter space, the probability that a random direction falls inside the "uphill" cone is below 10^{-5000}—not "very small," but physically inaccessible. Supervised BPTT reaches 97% directional accuracy in 600 epochs by taking first-order-guided steps through parameter space.

4. **We propose the Physical Necessity Hypothesis.** A set of physical constraints—binocular vision input, continuous-time dynamics, finite neuron budget, real-time motor output—defines a low-dimensional feasible solution space. Our GRU and the vertebrate tectum both settled into the same attractor. We propose this reflects physical necessity, not coincidence. The hypothesis makes a specific, falsifiable prediction: substituting the encoder (DINOv2 $\rightarrow$ CLIP $\rightarrow$ a CNN trained from scratch) or the

recurrent architecture (GRU → LSTM → continuous-time RNN) should reproduce the same structural convergence. If it does not, the hypothesis is falsified. These substitution experiments have not yet been performed—the hypothesis currently rests on a single data point (§6.3).

5. **Reverse validation through targeted lesioning.** Systematic damage to ten characterized circuit loci—spanning sensory ($\mathbf{W}_h$), gain-control ($\mathbf{b}_z$), recurrent ($\mathbf{U}_z$), motor-output ($\mathbf{W}_2$), and population-readout pathways—produces behavioral phenotypes that match the clinical presentation of corresponding biological lesions. A continuous bradykinesia-to-hyperkinesia spectrum emerges from $\mathbf{b}_z$ shift; unilateral $\mathbf{W}_2$ row ablation recapitulates contralateral saccadic deficits; sensory-column perturbation at δ/θ frequencies produces direction-specific resonance with $10\times$ amplification over random noise. Phenotypes are traceable to known circuit functions rather than post-hoc analogy. Eight of ten lesion sites produce graded, quantifiable deficits; the remaining two reveal structurally silent pathways ($\mathbf{U}_z \approx 0$, $\mathbf{U}_r \approx 0$), confirming that recurrent gating is eliminated by training rather than merely unused. (§5, Appendix E).

Appendix navigation. Seven appendices provide supplementary material: DINOv2 retinal signal properties (Appendix A), architecture evolution and failure record (Appendix B), physical limits of vertical avoidance (Appendix C), visual architecture optimization (Appendix D), lesion experiment protocols and complete data (Appendix E), complete parameter table (Appendix F), and code availability (Appendix G).

2 Architecture

Our system comprises three tiers. A frozen retina (DINOv2) converts stereo image pairs into a 1408-dimensional feature vector. A trainable midbrain (a 128-unit GRU) maps this vector to four thrust outputs. A body (point-mass kinematics) converts thrust into motion in a continuous toroidal space.

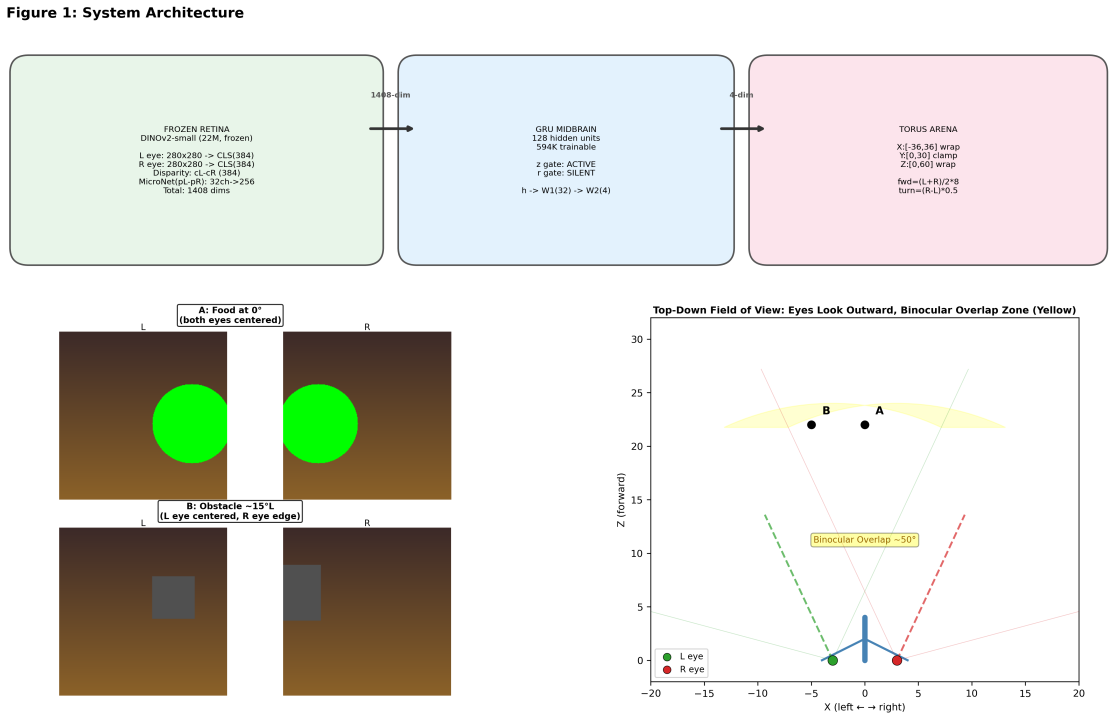

Figure 1: **System architecture. Top:** The visuomotor pipeline—frozen DINOv2 retina (1408-dim) to GRU(128) midbrain to 4-dim motor output in a torus arena. **Bottom left:** Two binocular-view pairs (A: food at 0°, both eyes centered; B: obstacle at ∼15° left, asymmetric eye views). **Bottom right:** Top-down field-of-view diagram showing eyes angled outward at ±25° and binocular overlap zone (∼50°).

2.1 Environment

The fish inhabits a three-dimensional toroidal arena: $X \in [-36, 36]$, $Y \in [0, 30]$, $Z \in [0, 60]$ in simulation units. X and Z boundaries torus-wrap (crossing one edge instantly reappears at the opposite edge), eliminating wall collisions as a confound. Y boundaries are soft-clamped—the fish cannot leave the water column, but ascending or descending incurs no penalty. Food appears as green textured spheres (radius 3–8); obstacles as gray rectangular boxes. Rendering uses a pinhole camera model with focal length fl $= 80$, image size 280×280 pixels, field of view $\approx 100°$ horizontal $\times$ $100°$ vertical. The two eyes are separated by ± 3.0 units on the horizontal axis and canted outward by 25° from the forward direction, producing a binocular overlap zone of $\sim 50°$ (see Appendix D). Both eyes share the same Y-axis coordinate, with zero vertical offset (making vertical disparity inaccessible; discussed in Appendix C).

We chose the torus as the minimal nontrivial geometry—allowing the fish to swim indefinitely without getting lost, guaranteeing obstacle presence, and eliminating "turn away from the wall" as a confound for avoidance strategy.

2.2 Retina: A Frozen DINOv2-small

We use DINOv2-small [35], a Vision Transformer (ViT-S/14) pretrained in self-supervised fashion on 142 million images collected from the internet. Its 22M parameters remain frozen throughout. Why a frozen visual encoder? Because the architectural premise is: *a sufficiently rich, task-agnostic visual representation already contains the spatial-structural* *information needed to inform behavior.* If this is correct, then training a dedicated vision encoder for the motor task is over-engineering.

The system renders left-eye and right-eye views. Each 280×280 image is resized to 224×224 (DINOv2's expected input) and forwarded independently through DINOv2, yielding:

- **CLS tokens** (384-dim $\times$ 2): global scene descriptors, L2-normalized to the unit sphere
- **Patch tokens** (256 patches $\times$ 384-dim $\times$ 2): local spatial features, L2-normalized

The visual input vector has dimension 1408:

$$[\ \mathbf{c}^L_{(384)}\ \mid\ \mathbf{c}^R_{(384)}\ \mid\ (\mathbf{c}^L - \mathbf{c}^R)_{(384)}\ \mid\ \text{MicroNet}(\mathbf{p}^L - \mathbf{p}^R)_{(256)}\ ] \tag{1}$$

The disparity channel $\mathbf{c}^L - \mathbf{c}^R$ captures binocular differences—the distinct offset at which an object appears in each eye, the foundation of depth inference. MicroNet is a tiny two-layer MLP ($16 \times 384 \to 8 \times 16 \to 256$; $\sim$10K parameters, trained in early experiments and then frozen as a fixed-function subnetwork) that processes patch differences across 32 spatial regions into a learned, spatialized disparity map—analogous to the developmental locking of activity-dependent functional specialization in the biological retina.

DINOv2 has never seen "food" or "obstacle." It has zero concept of green spheres and gray boxes—only the structural relationships it learned across billions of image patches. A systematic analysis of DINOv2 signal properties in Appendix A establishes that normalized CLS tokens and patch differences carry azimuthal information sufficient to support depth inference and object discrimination.

2.3 Midbrain: GRU(128)

We refer to the trainable midbrain as **Tecto**. It is a Gated Recurrent Unit (GRU) with 128 hidden units [36] that maps the 1408-dimensional visual input into a 128-dimensional hidden state $\mathbf{h} \in \mathbb{R}^{128}$, then decodes it into a 4-dimensional motor output $[L, R, U, D] \in [-1, +1]^4$. All outputs are mathematically symmetric: L and R jointly contribute forward thrust $(L + R)/2$ and turning torque $(R - L)$; U and D control vertical motion.

The standard GRU equations are:

$$\mathbf{z} = \sigma(\mathbf{W}_z\mathbf{x} + \mathbf{U}_z\mathbf{h} + \mathbf{b}_z) \tag{2}$$
$$\mathbf{r} = \sigma(\mathbf{W}_r\mathbf{x} + \mathbf{U}_r\mathbf{h} + \mathbf{b}_r) \tag{3}$$
$$\tilde{\mathbf{h}} = \tanh(\mathbf{W}_h\mathbf{x} + (\mathbf{r} \odot \mathbf{h})\mathbf{U}_h + \mathbf{b}_h) \tag{4}$$
$$\mathbf{h}' = (1 - \mathbf{z}) \odot \mathbf{h} + \mathbf{z} \odot \tilde{\mathbf{h}} \tag{5}$$
$$\mathbf{h}' \leftarrow 0.999 \cdot \mathbf{h}' \quad \text{(non-standard decay, added for numerical stability)} \tag{6}$$

Output pathway: $\tanh\left(\text{ReLU}(\mathbf{h}'\mathbf{W}_1^T + \mathbf{b}_1)\mathbf{W}_2^T + \mathbf{b}_2\right) \to [L, R, U, D]$, where $\mathbf{W}_1 \in \mathbb{R}^{32\times128}$, $\mathbf{W}_2 \in \mathbb{R}^{4\times32}$.

The hidden dimension $H = 128$ was chosen through systematic reduction: three independent runs confirmed $H = 64$ as a capacity bottleneck; $H = 128$ provides a balance between gating fidelity and parameter efficiency.

Total trainable parameters: 594,468. Of these, approximately 397K are actively used (§4.4). The remaining 197K—$\mathbf{W}_r$, $\mathbf{U}_r$, $\mathbf{b}_r$—are structurally silent because r is identically 0.5 under all conditions (§4.1).

Kinematic equations:

$$\text{forward speed} = \frac{L+R}{2}\times 8\,\text{units/step}, \quad \text{turn rate} = (R-L)\times 0.5\,\text{rad/step}, \quad \text{vertical speed} = (U-D)\times \tag{7}$$

Maximum forward speed is 8 units/step, maximum turn rate is 1 rad/step. These values were empirically tuned to produce smooth, "fish-like" motion in animations, not calibrated from any biological measurement.

2.4 Training: Heuristic Supervision via MSE

The training paradigm follows a heuristic controller: a simple algorithm that independently produces food-seeking and obstacle-avoiding thrust vectors through distance-weighted fusion. For each food-obstacle configuration drawn from a minimal visible set (1 food, 1–4 obstacles per sample; 10% empty to debias), the oracle computes target thrusts $[L, R, U, D]_{\text{target}}$. The GRU is trained via mean squared error to match this target, using per-sample single-pass forward propagation (no unrolled BPTT through time).

- Training data: 24,000 independent random samples
- Loss: $\text{MSE}(\text{output}, \text{oracle})$
- Optimizer: Adam, $\text{lr} = 0.001$
- LR schedule: ReduceLROnPlateau, patience=40, factor=0.5
- Epochs: 600
- Initial weights: Xavier uniform initialization (zero initialization prevents gating from functioning)
- Final validation loss: 0.018705

Two key choices merit attention. First, **independent random samples**—rather than trajectory sequences—sample the azimuthal distribution uniformly in each batch, forcing the GRU to develop visual input utilization rather than learning biases in the training distribution. Trajectory data is biased toward forward-facing objects and produces lazy policies (swim forward; hope to encounter food).

Second, **Xavier initialization**—rather than zero or small random numbers—unlocks the gating mechanism: z and r under zero-initialization output constant 0.5, meaning the hidden state never changes, gradients are zero, and the network never learns. This is a general lesson: recurrent networks require non-zero activations for their gating mechanisms to function—unlike feedforward networks, where zero-initialized weights and gradually increasing activations are standard.

3 Behavioral Baselines

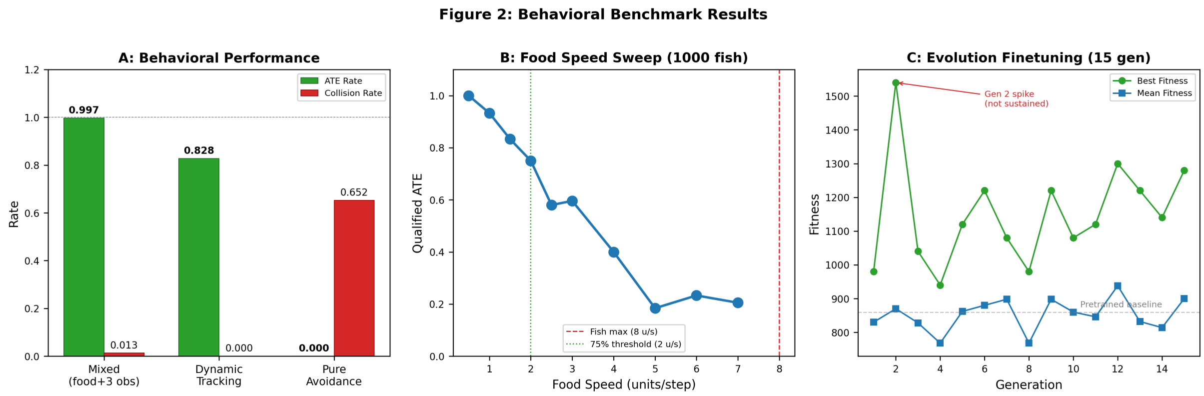

Figure 2: **Behavioral benchmark results. A:** ATE and collision rates across mixed, dynamic tracking, and pure avoidance conditions ($n = 1550$ total fish). **B:** Qualified ATE vs. food speed (10 levels $\times$ 100 fish; smooth degradation confirms population coding). **C:** Fifteen generations of evolution finetuning show no systematic improvement (mean fitness slope $+2.7 \pm 2.9$/gen, $p = 0.35$).

3.1 Static Food + Obstacles: Near-Perfect Within Distribution

We tested 300 fish, each for 400 steps, with 1 static food item and 3 random obstacles. Food position is guaranteed visible at spawn. The primary metric is **ATE** (Ate The Food): a binary success flag indicating the fish captured the food item (distance < food_radius+2 at any step). Collisions with obstacles incur no energy penalty—we simply count contact events.

Table 1: Cross-tabulation of ATE and collision outcomes for 300 fish in the mixed environment (1 food + 3 obstacles, 400 steps). All 4 fish that collided also ate—collision did not prevent task completion.

	Collision	**No Collision**	**Total**
ATE!	4	295	299
No ATE	0	1	1
Total	4	296	300

These results approach the theoretical limit of the defined task. Within the training distribution, the trained brain captures food reliably while avoiding obstacles. On unseen obstacle configurations, one fish fails, four graze an obstacle, and 295 navigate without incident.

The critical result is that the GRU's visuomotor mapping accurately reproduces the oracle's targeting behavior: trained purely on static MSE regression against the heuristic controller's thrust vectors, the network achieves near-perfect food capture without ever interacting with the environment during training.

3.2 Tracking Randomly-Walking Food: Generalization Without Training

We did not train the GRU on dynamic food. The oracle provides static target thrusts. To test whether the population-code hypothesis implies generalization to dynamic targets, we generated 500 fish with a single randomly-walking food item (speed fluctuating 0.5–2.5 units/step, heading subject to random drift), no obstacles. A "qualified" fish is one whose food is in its field of view at spawn.

Table 2: Dynamic food-tracking benchmark (500 fish, one randomly-walking food, 400 steps, no obstacles).

Metric	**Value**
Total fish	500
Qualified (saw food at spawn)	215 (43.0%)
Qualified ATE	**178/215 = 82.8%**
All-fish ATE	188/500 = 37.6%
Mean steps to ATE	6.6
Misses (qualified, no capture)	37 fish; mean min. distance 20.6

Among fish that could see the food at the starting instant, 83% captured the target—with zero training on moving targets. This is not incidental hits. The 37 misses *got close* (mean minimum distance 20.6) but not within 7 units—suggesting execution failure ("chased but didn't catch"), not perceptual failure ("didn't see"). The mapping from motor dynamics to population code generalizes smoothly: when food moves, the same tuning curves re-weight to a new azimuthal vector.

The majority of the population did not see the food at spawn (285/500 = 57%), reflecting that food is not always within the field of view under our single random-spawn configuration. When a fish cannot see the food, it cannot—by definition—track it; it adopts the same "idle" cruising state ($L \approx R \approx 0.53$, undirected forward motion) observed in torus testing.

3.3 Food-Speed Sweep: Tracking Degradation Is Smooth—Not a Crash

To quantify tolerance to velocity, we scanned food speed from 0.5 to 7.0 units/step (fish maximum 8) across 10 speed levels with 100 fish per level.

Table 3: Food-speed sweep (10 levels $\times$ 100 fish = 1000 fish).

Food speed (u/s)	% of fish max speed	Qualified ATE
0.5	6.3%	**1.000**
1.0	12.5%	**0.933**
1.5	18.8%	**0.833**
2.0	25.0%	**0.750**
2.5	31.3%	0.580
3.0	37.5%	0.596
4.0	50.0%	0.400
5.0	62.5%	0.184
6.0	75.0%	0.233
7.0	87.5%	0.205

Degradation is smooth—with a shoulder around 50% of fish maximum speed at 2.0—and the residual capture rate at 5.0–7.0 is approximately 20%, consistent with pure-chance encounter rate (rough estimate).

This smooth degradation is the signature of a population code. A categorical strategy ("food on the left: hard left turn") would produce a sharp drop-off at some threshold speed; instead we see a gradual decline roughly linear in the food-to-fish speed ratio—suggesting each hidden neuron casts a vote for a dynamic target, and the population average loses lock as the target accelerates. Population codes naturally produce smooth interpolation, not only across spatial positions, but across the kinematic continuum of moving targets.

3.4 Pure Obstacle Avoidance: Behavior Vanishes Without Food

As a control, we tested 500 fish with no food whatsoever and 15 random obstacles—15 was chosen to perceptually overwhelm the fish. All collision metrics reported here reflect unavoidable contact in an environment saturated with obstacles, not a failure of the visuomotor controller.

Table 4: Pure obstacle avoidance: 500 fish, 15 obstacles, zero food. Collision rate is independent of spawn-time obstacle visibility. No fish survived 400 steps without collision.

Metric	Value
Fish that collided ($\geq$1 hit)	326/500 (65.2%)
Collision-free	174/500 (34.8%)
Survived 400 steps, no collision	0/500 (0.0%)
Mean steps to first collision	13.8 (median: 11.0)
Mean collisions per colliding fish	2.67
Collision rate when 0 obs. visible at spawn	64.3%
Collision rate when 15 obs. visible at spawn	75.0%

With food (§3.1), the collision rate is 1.3%. Without food, it is 65.2%—a fifty-fold difference that reveals obstacle avoidance as food-driven, not autonomous.

The GRU requires the presence of food to anchor its behavior. When food is visible, obstacles manifest as perturbations on a foraging baseline ($\cos(\text{mixed}, \text{food}) = 0.914$;

§4.2). When there is no food to define a baseline "correct direction," the hidden state wanders—a mix of undirected swimming and incidental obstacle contact. Collision rate is approximately 65%, regardless of the number of obstacles visible at spawn. The collision rate does not depend on whether 0 or 15 obstacles are visible. Without a food-driven anchor for the behavioral model, the GRU does not "see" obstacles.

This has direct implications for understanding multitasking in biology. The tectum does not process foraging and avoidance independently. Avoidance operates as "feeding minus an obstacle perturbation"—a competitive mechanism that rides on top of a foraging baseline. When no foraging motivation exists, the competitive map degenerates to an undirected baseline.

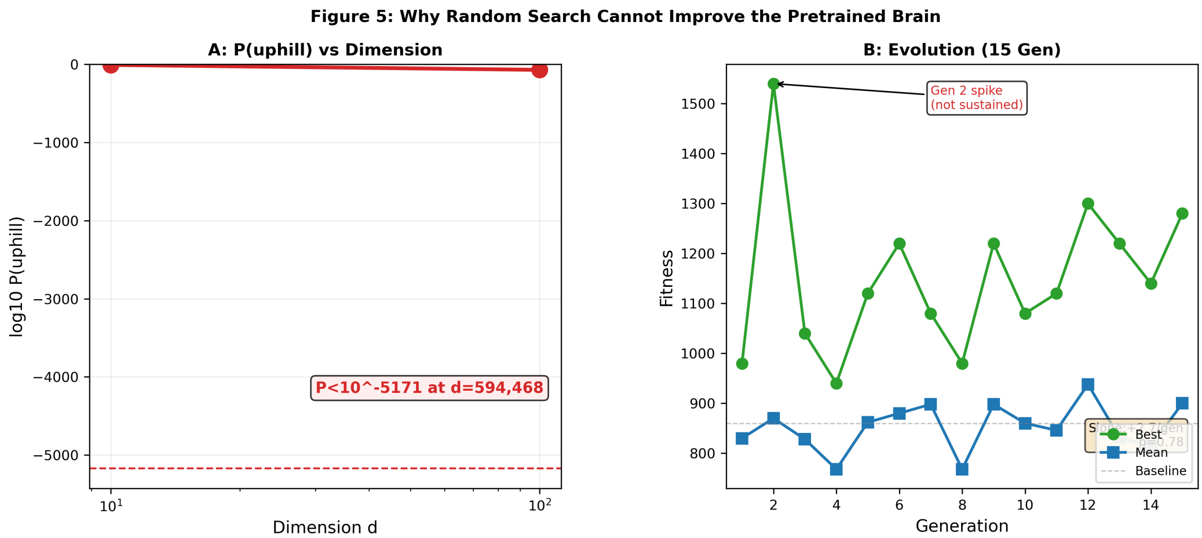

Figure 3: **Why random search cannot improve the pretrained brain. A:** Log-probability that a random mutation points uphill vs. dimension d. At $d = 594{,}468$, $P < 10^{-5171}$—physically inaccessible. **B:** Fifteen generations of evolution: best-fitness spikes are transient; mean fitness shows no trend ($+2.7 \pm 2.9$/gen, $p = 0.35$).

3.5 Evolution Finetuning Cannot Improve the Pretrained Brain

If the pretrained brain sits at a deep local minimum in the 594K-parameter space, then random mutations—even paired with tournament selection—should produce no net improvement. This is exactly what we found.

We seeded a population of 10 fish with the pretrained brain, added small Gaussian noise (σ = 50% of the mutation scale), and evolved for 15 generations under energy-metabolic pressure (3 episodes of 150 steps per fish per generation; energy 150 $\rightarrow$ basal -0.8, collision -10, food $+60$, cap 250).

Table 5: Fifteen generations of evolution finetuning starting from the pretrained brain. Elite 25%, tournament selection (3-way), mutation rate 3% per weight, mutation scale 10% of pretrained weight standard deviation.

Generation	**bestF**	**meanF**	**worstF**
1	980	830	640
2	**1540**	870	600
6	1220	880	640
12	1300	938	660
15	1280	900	560

The peak at Generation 2 (bestF=1540) is a classic lucky-mutation spike—a favorable random configuration assembled in one individual, lost to tournament drift in the next generation. Elite preservation (25%) is helpless against the mutation rate: ∼17,834 weights are replaced per fish per generation, meaning a beneficial mutation is diluted at the next crossover event.

Using linear regression (excluding the Gen-2 outlier), the slope of meanF is $+2.7 \pm 2.9$ per generation ($p = 0.35$), consistent with no improvement over 15 generations.

The inefficacy of evolution finetuning has a geometric explanation. In a 594,468-dimensional space, the expected angle between a random direction and the true gradient (the "uphill" direction) is 90°—because almost all the area on a high-dimensional unit sphere is concentrated near the equator. The probability that a random direction falls within 10° of the true gradient is:

$$P(\cos\theta > 0.98) \approx (1 - 0.98^2)^{d/2} \approx (0.0396)^{297234} \approx 4.7 \times 10^{-5171} \tag{8}$$

This is 10^{-5000}. For a flat parameterization of 594K independent weights with no hierarchical structure, an uphill mutation is physically inaccessible. Supervised BPTT accumulated approximately $24{,}000 \times 600 = 14{,}400{,}000$ precise gradient steps through this same space—each step pointing toward the true downhill direction. The efficiency gap between evolution and supervision is not a small multiple—it is *geometric*, spanning a complexity difference on the order of $d/\ln(d)$, which, for $d \approx 6 \times 10^5$, is astronomically large.

Why, then, can biology evolve a tectum? Because biology does not search through a flat synaptic weight space. A single Hox gene mutation restructures thousands of synapses through a program that is itself low-dimensional—genome → gene regulatory networks → developmental rules → neuronal connectivity. This hierarchical compression reduces the effective search dimensionality by orders of magnitude, making uphill directions geometrically accessible. Our GRU lacks this compression: its 594K parameters are independent. Hence it requires gradients. The 10^{-5000} argument is not a claim about biology—it is a claim about the unsuitability of flat random search as a training algorithm for high-dimensional continuous parameter spaces.

The full cost analysis and geometry of random search are developed in the supplementary material; also see [37] for the general geometry of high-dimensional loss landscapes.

4 Circuit Analysis

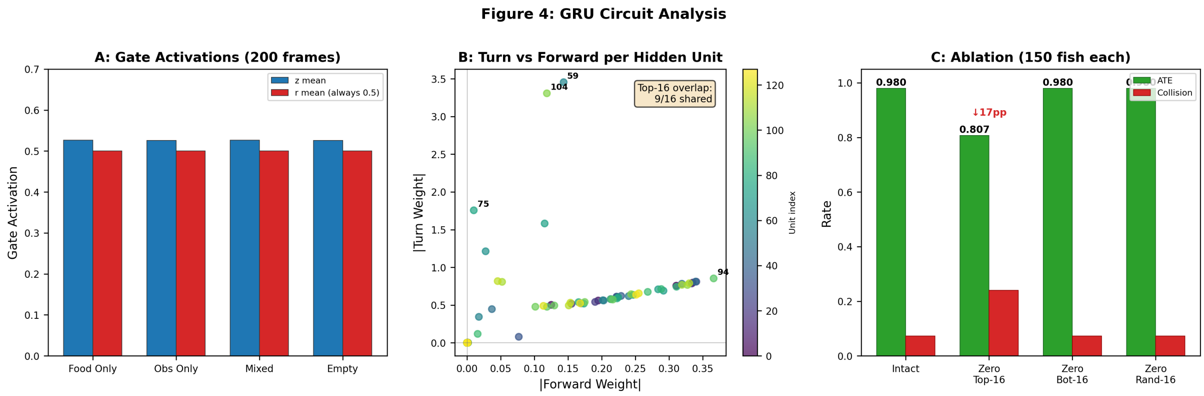

Figure 4: **GRU circuit analysis summary. A:** Gate activations across four scenarios (200 frames). The update gate z varies modestly (~0.526); the reset gate r is fixed at 0.500 ($\sigma = 0$). **B:** Per-hidden-unit effective weight contributions to turn vs. forward thrust ($\mathbf{W}_{\text{eff}} = \mathbf{W}_2 \times \mathbf{W}_1$). Top-16 turn and top-16 forward units overlap 9/16—no dedicated "turn neurons." **C:** Ablation results (150 fish per condition). Zeroing the top-16 units reduces ATE from 0.980 to 0.807 (−17 pp) and raises collision rate from 7.3% to 24.0% (+16.7 pp, a 3.3× increase); zeroing the bottom-16 or random-16 has zero effect.

If this network performs this well, *how* does it do it—every mechanism stepping from visual input through gating through hidden state through output? We conducted systematic circuit analysis in four phases: gate probing (§4.1), hidden manifold (§4.2), weight-pathway tracing (§4.3), and ablation (§4.4). What emerged: the network's organization is as much *discovered* as it is designed—the loss function specified what to optimize; gradient descent over 600 epochs determined the circuit-level solution.

4.1 Finding One: The Reset Gate r Is Dead

We generated 200 frames across four scenarios (food-only, obstacle-only, mixed, empty) and probed the 128-dimensional activations of the update gate z and reset gate r. For each scenario, we recorded gate statistics, outputs, and scene variables.

Table 6: GRU gate-state statistics across four scenarios (200 frames).

Scenario	$z_{\textbf{mean}}$	$r_{\textbf{mean}}$	$r_{\textbf{std}}$
food_only	0.5263	0.5000	0.0000
obs_only	0.5261	0.5000	0.0000
mixed	0.5264	0.5000	0.0000
empty	0.5256	0.5000	0.0000

The reset gate r has zero standard deviation across all scenarios, all frames. $r = 0.5$ is exactly $\sigma(0)$, meaning the sum $\mathbf{W}_r\mathbf{x} + \mathbf{U}_r\mathbf{h} + \mathbf{b}_r$ never deviates from zero. The associated 197K parameters ($\mathbf{W}_r$, $\mathbf{U}_r$, $\mathbf{b}_r$) are never used by the trained network.

The effective GRU formula simplifies to:

$$\tilde{\mathbf{h}} = \tanh(\mathbf{W}_h\mathbf{x} + 0.5 \cdot \mathbf{h}\mathbf{U}_h + \mathbf{b}_h) \qquad (r \text{ does not gate}) \tag{9}$$

$$\mathbf{h}' = (1 - \mathbf{z}) \odot \mathbf{h} + \mathbf{z} \odot \tilde{\mathbf{h}} \qquad (\text{only } z \text{ is active}) \tag{10}$$

The update gate z carries behaviorally-relevant signal. z_{mean} correlates with forward thrust ($r = +0.52$) and with turn magnitude ($r = +0.47$)—but z does *not* distinguish between scenarios. z does not encode "this is food" versus "this is an obstacle." It encodes *behavioral intensity*—"how fast to move"—not *behavioral type*.

This directly parallels cholinergic gain control of the tectum. Brainstem cholinergic projections from the pedunculopontine and laterodorsal tegmental nuclei provide the primary neuromodulatory input to the superior colliculus, regulating the responsiveness of tectal motor neurons without carrying spatial or object-level information. The z gate does the same thing: it sets how strongly the network responds to sensory input, not where it directs the response. When the fish moves faster, z rises (more receptive to new information); when it cruises, z hovers near 0.5 (conservative exponential moving average). This cholinergic gain-control interpretation is the same mechanism underlying the bradykinesia-to- hyperkinesia spectrum revealed by $\mathbf{b}_z$ shift (§5): dopamine loss in Parkinson's disease reduces cholinergic drive to the tectum, effectively lowering z.

The redundancy of these 197K parameters is evidence of architectural over-provision. The standard GRU has three gating matrix sets. A continuous visuomotor pipeline needs only one—exponential smoothing suffices. Complex gating mechanisms like the LSTM's forget gate are designed for discrete events and long-range dependencies—but in a "see → move" pipeline, they are superfluous.

4.2 Finding Two: Food-Seeking and Obstacle-Avoidance Are Orthogonal in Hidden Space

We collected 2,700 frames of hidden-state vectors (45 trajectories × 60 steps, divided into three scenarios: food-only, obstacle-only, mixed) and ran PCA on the resulting 2700×128 matrix.

PCA explained variance: PC1 = 57.0%, PC2 = 33.2% (first two PCs capture 90.2%). **Critically, the cosine similarity between mean hidden states across scenarios:**

Table 7: Mean hidden-state cosine similarity across scenarios.

	food_only	**obs_only**	**mixed**
food_only	1.000	**−0.047**	0.914
obs_only	−0.047	1.000	0.249
mixed	0.914	0.249	1.000

When only food is present, the hidden state occupies one direction. When only obstacles are present, it occupies the nearly *opposite* direction ($\cos \approx -0.047 \approx 90°$ angle). When both food and obstacles are present, the hidden state is approximately the food direction plus a small, negatively-weighted obstacle perturbation: $\mathbf{h}_{\text{mixed}} \approx \mathbf{h}_{\text{food}} - w \cdot \mathbf{h}_{\text{obs\_perturbation}}$, with $\cos(\text{mixed}, \text{food}) = 0.914$.

This *is* the competitive sensory map of the tectum. Biologically, the tectum's superficial layers receive direct retinotopic input; its deep layers contain a motor map where multiple sensory signals converge onto the same population of premotor neurons and compete through local inhibitory interneurons embedded within the tectum itself. The competition between excitatory (food-seeking) and inhibitory (obstacle-avoidance) signals is resolved in parallel on a single map. Our GRU implements this competition entirely within its 128-dimensional hidden state, with no external module—the obstacle signal is a perturbation vector subtracted from the food direction in the same population, producing a unified motor output without switching between modes.

Why orthogonality is efficient. If food and obstacle representations shared a subspace, they would interfere: approaching food would also amplify the obstacle signal, a dangerous cross-contamination. Orthogonality means the obstacle signal never dilutes the food signal—they merely cast separate shadows on different axes of the latent space. The mixed-scenario hidden state can therefore be a unified representation $\mathbf{h}_{\text{eat}} + \mathbf{h}_{\text{avoid}}$ without needing to "switch" between modes.

Note: only 18 frames were behaviorally labeled as "avoiding obstacle"—reflecting that, in pure-obstacle scenarios, strong avoidance maneuvers are scarce. Most obstacle frames are "near-obs"—passively close to an obstacle without active evasion. This matches the training: the heuristic controller only triggers avoidance when an obstacle is genuinely threatening, not preemptively. The GRU learned the same strategy.

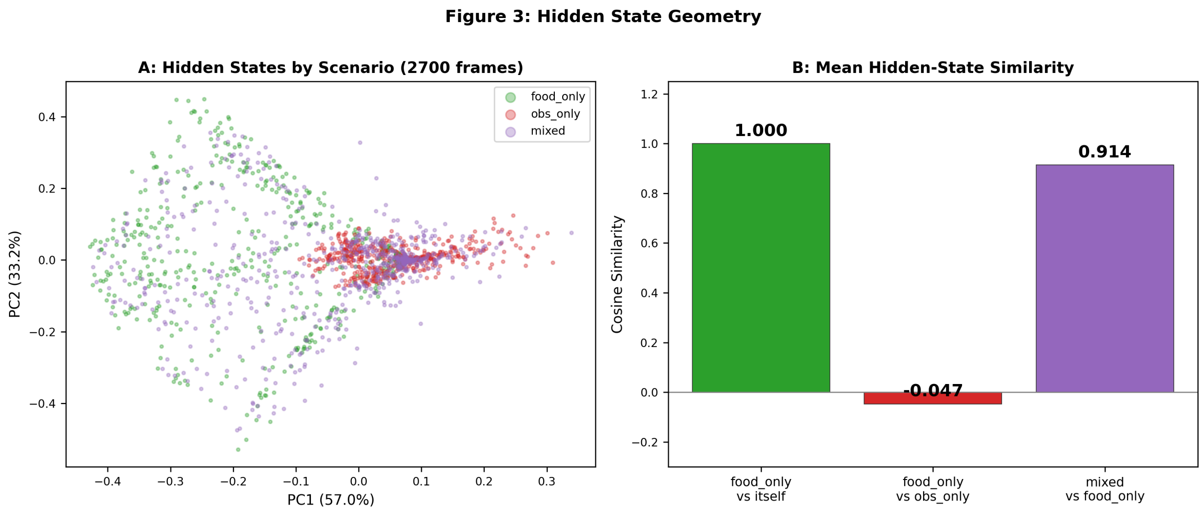

Figure 5: **Hidden state geometry (PCA of 2700 trajectory frames).** **A:** Hidden states colored by scenario. Food-only and obstacle-only states form clearly separated clusters (cosine similarity -0.047). Mixed-scenario states cluster near food-only (cos 0.914). **B:** Mean hidden-state cosine similarity, confirming near-orthogonal encoding and mixed $\approx$ food $-w\cdot$ obs.

4.3 Finding Three: Behavioral Signals Are Population-Coded—No Dedicated Neurons

If the GRU had a dedicated "left-turn neuron" and "right-turn neuron," they should appear in the effective output-weight matrix. We computed the hidden-dimension-to-output mapping:

$$\mathbf{W}_{\text{eff}} = \mathbf{W}_2(4 \times 32) \ \times \ \mathbf{W}_1(32 \times 128) = (4 \times 128) \tag{11}$$

This gives a 4×128 matrix where each column (each hidden dimension) contains 4 weights for that dimension's contribution to the four outputs.

Table 8: Top hidden dimensions ranked by |turn_w| and |fwd_w|. Overlap between top-16 |turn| and top-16 |fwd| is 9/16. No dimension is "pure" turn or "pure" forward.

Rank	**Dim**	**|turn_w|**	**|fwd_w|**	**Ratio**
1	59	3.455	0.143	24.1
2	104	3.307	0.119	27.9
3	75	1.756	0.010	175.7
4	64	1.213	0.028	44.0
5	115	0.818	0.046	18.0
6	111	0.808	0.052	15.5
7	73	1.582	0.115	13.7
⋮	⋮	⋮	⋮	⋮
128	61	∼0	∼0	—

The top |turn| dimension (dim 59: +3.455, pushing left turn) and the top opposite-turn dimension (dim 104: −3.307, pushing right turn) also contribute positively to forward thrust. The top-16 |turn| dimensions and top-16 |fwd| dimensions share 9/16 dimensions. **There are no turn-dedicated neurons. There are no forward-dedicated neurons.** The tuning is thoroughly population-wide: to produce "turn left," you need coordinated changes across nearly all 128 hidden dimensions—individual dimensions contribute only small net components that sum with all others.

This is the organization of tectal motor neurons. A single tectal neuron has a movement field—it fires for a range of directions but peaks at a preferred saccade direction. To produce a *specific* saccade, the tectum requires population-vector decoding—each neuron votes with its preferred direction, and the weighted average determines the final motor command. Our $\mathbf{W}_2$ matrix does exactly this: output = $\tanh(\mathrm{ReLU}(\mathbf{h}\mathbf{W}_1^T + \mathbf{b}_1)\mathbf{W}_2^T + \mathbf{b}_2)$ is a linear vectorized readout translating the 128-dimensional population activity into a 4-dimensional motor command.

4.4 Finding Four: ∼12.5% of Hidden Dimensions Are Silent Synapses

What happens if we compute each hidden dimension's effective weight ($|\mathbf{W}_{\text{eff}}|$ summed row-wise), rank by importance, then ablate (zero out) the top-16, bottom-16, and random-16?

Table 9: Ablation results: 150 fish per condition, 3 obstacles + 1 food.

Condition	**ATE**	**Collision Rate**
Intact (no ablation)	98.0%	7.3%
Zero top-16	**80.7%**	**24.0%**
Zero bottom-16	98.0%	7.3%
Zero random-16	98.0%	7.3%

The bottom-16 dimensions have $|\mathbf{W}_{\text{eff}}| < 0.0003$—essentially zero. Ablating them has no effect. They are the equivalent of **silent synapses** in the tectum: connections physically present ($\mathbf{W}_1$ has non-zero entries), but no AMPA receptors mean they carry no current ($\mathbf{W}_2$'s readout weight is zero). Whether these silent dimensions can be recruited for functional recovery after injury—as biological silent synapses are during rehabilitation—has not been tested and would require a plasticity experiment (e.g., retraining the ablated network and measuring whether formerly silent units become active).

Ablating the top-16 dimensions causes measurable but not catastrophic degradation: ATE drops 17.3 percentage points (from 98.0% to 80.7%), collision rate rises 16.7 percentage points (from 7.3% to 24.0%, a 3.3× increase). The remaining ∼112 effective dimensions partially compensate—a demonstration of population-code redundancy. If "left turn" were encoded by a single dedicated neuron, killing that neuron would abolish left turning. Population code: killing the 16 biggest voters changes the vote by 17 percentage points.

Efficiency. Of 594,468 trainable parameters, approximately 397K are actively used (67%). The dead weights correspond to $\mathbf{W}_r$, $\mathbf{U}_r$, $\mathbf{b}_r$ (∼197K; §4.1) plus 16 zero columns in $\mathbf{W}_1$ (∼512 weights). These are not bugs—they are the model's implicit regularization. $\mathbf{W}_1$ learned during training which hidden dimensions are worth reading out and which are not. The dimensions that got pruned are those that provide signals highly redundant with existing readout—$\mathbf{W}_1$ has no incentive to keep them.

5 Lesion Validation

The preceding sections established Tecto's structural convergence to the optic tectum through forward characterization in §4.1–§4.4: gate analysis revealed z as gain control and r as silent, hidden-state analysis revealed orthogonal food/obstacle coding, weight-pathway tracing revealed population-vector decoding, and ablation revealed silent synapses. This section builds directly on that circuit characterization to provide reverse validation: each lesion targets a component whose function was established in §4, and the resulting behavioral deficit should trace cleanly to the damaged mechanism. We systematically damaged ten circuit loci spanning sensory ($\mathbf{W}_h$), gain-control ($\mathbf{b}_z$), motor-output ($\mathbf{W}_2$), population-readout (hidden-unit ablation), and recurrent-feedback ($\mathbf{U}_z$, delayed $\mathbf{W}_h^T$) pathways. Every lesion maps to a known computational function characterized in §4 and every phenotype traces to a specific circuit-level damage. Full protocols and data tables are in Appendix E.

We report at three confidence levels: **measured** (directly traceable to experimental output), **interpreted** (inferred with circuit-level reasoning), and **speculative** (clinical correspondence requiring independent biological validation).

5.1 Sensory Pathway: Direction-Specific Frequency Resonance

Dim 59 of $\mathbf{W}_h$ carries the strongest left-turn weight in the circuit (§4.3). We injected a sinusoidal perturbation $\mathbf{p} = \varepsilon \sin(2\pi f t)\, \mathbf{w}_h^{59}$ into the visual input, scanning $f \in [1, 30]$ Hz. We tested three control vectors identically: a random unit vector, a $\mathbf{W}_z$ column, and a weakly-weighted $\mathbf{W}_h$ column.

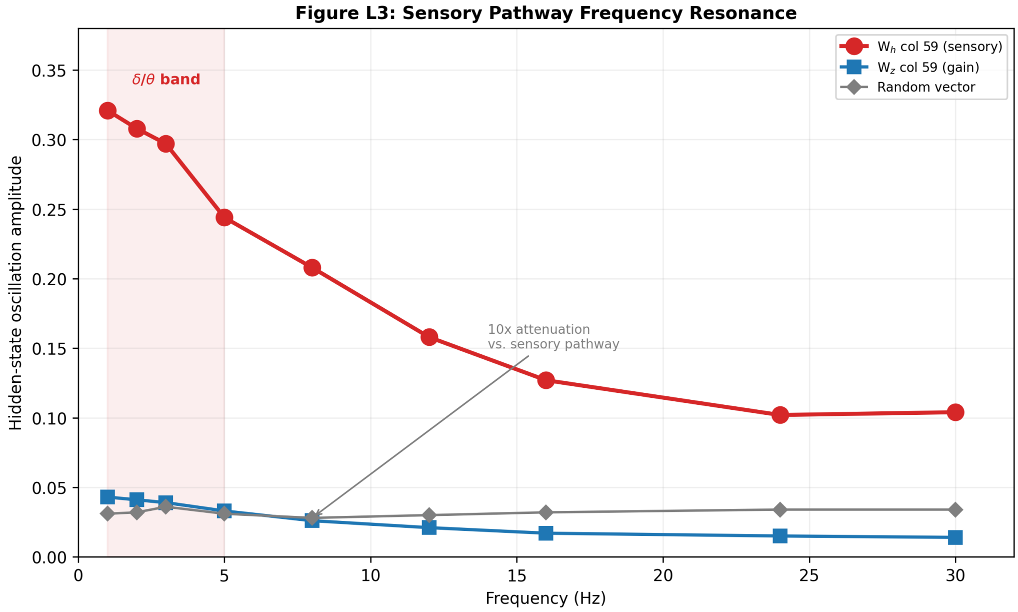

Figure 6: **Frequency response of sensory vs. gain pathways.** The $\mathbf{W}_h$ sensory column produces oscillation amplitudes 10× higher than random perturbation in the δ/θ band (1–5 Hz), decaying smoothly above 8 Hz—a first-order low-pass filter. $\mathbf{W}_z$ and random vectors show flat, attenuated responses across all frequencies.

Measured. The $\mathbf{W}_h$ column produces $h_{\text{osc}} \approx 0.30$–$0.32$ in the 1–5 Hz band, decaying to ∼0.10 at 30 Hz (Fig. 6). Random and $\mathbf{W}_z$ vectors produce $h_{\text{osc}} \leq 0.04$ at all frequencies—attenuation exceeding 10×.

Interpreted. The $\mathbf{W}_h$ pathway acts as a first-order low-pass filter with corner frequency ∼3–5 Hz, determined by the GRU's intrinsic time constants. Direction-specific perturbations resonate because they match the pathway's natural transfer function.

Speculative—clinical correspondence. The δ/θ band (1–5 Hz) is the frequency range of absence and focal seizures. If a focal epileptic circuit develops within the superior colliculus and produces rhythmic discharge at δ/θ frequencies, the SC's motor output would be driven into directional oscillation biased toward the encoded direction of the affected neuronal population— matching the directional gaze deviation observed during ictal discharge. The computational basis is the direction-specific sensory channel identified here: rhythmic drive into a single sensory column produces amplified, direction-locked motor oscillation at the resonant frequency of the tectal circuit. **Open question:** whether this low-pass characteristic persists with a multi-synaptic feedback pathway rather than the single-column probe used here.

5.2 Gain Control: A Continuous Spectrum from Bradykinesia to Hyperkinesia

The update gate bias $\mathbf{b}_z$ sets baseline gain. Shifting it by Δ changes global responsiveness without altering directional tuning—consistent with the established role of z as encoding behavioral intensity (§4.1). We tested $\Delta \in [-0.50, +0.50]$ (30 fish per condition).

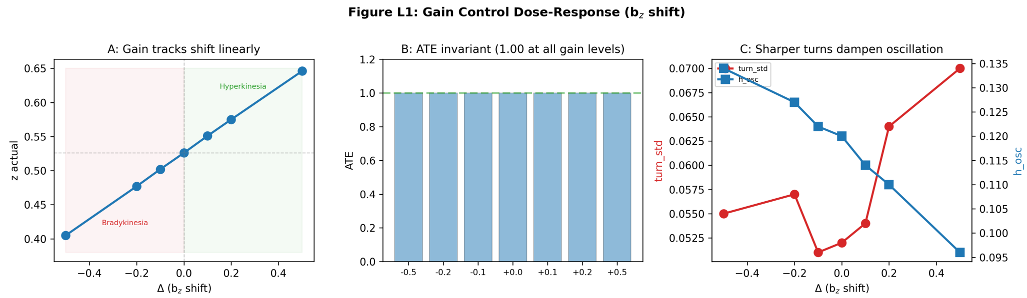

Figure 7: **Gain-control dose-response.** z_{actual} tracks Δ linearly (0.40–0.65). ATE remains at 1.00 throughout—task completable at all gain levels, movement style varies. Turn variability increases with gain while hidden-state oscillation decreases—higher z produces sharper turns that dampen internal oscillation.

Measured. z_{actual} tracks Δ linearly from 0.405 to 0.646. ATE remains 1.00. Turn_std rises from 0.055 to 0.070; h_osc declines from 0.134 to 0.096 (Fig. 7).

Interpreted. Gain forms a continuous spectrum, not a switch. The system remains functional across a wide range—motor *style* changes but task *competence* is preserved. The decline in h_{osc} with increasing z is a non-intuitive finding: conventional intuition suggests that higher gain should amplify oscillation. The opposite occurs because higher z produces sharper, more decisive individual turns—each turn dissipates directional energy in a single large correction rather than accumulating it as sustained oscillatory hidden-state dynamics. Low z, by contrast, generates many small, hesitant adjustments that never fully resolve the directional error, causing the hidden state to ring at the GRU's natural frequency. In dynamical-systems terms, high gain moves the system toward an overdamped regime where each perturbation is absorbed in a single step; low gain leaves the system underdamped, where perturbations decay slowly through multiple cycles of the recurrent dynamics. This gain-oscillation relationship mirrors the clinical observation that bradykinetic patients (low effective z) often exhibit tremor—a rhythmic oscillation at ∼4–6 Hz—while hyperkinetic patients exhibit rapid, non-oscillatory movements.

Speculative—clinical correspondence. $\Delta < 0$: bradykinetic, resembling early Parkinsonian hypokinesia (dopamine loss reduces cholinergic drive to tectum, lowering effective z). $\Delta > 0$: hyperkinetic, resembling Huntington's chorea (striatal GABA loss disinhibits gain modulation). Preserved ATE explains why early-stage movement disorders go undetected—task completion rate is a poor diagnostic metric.

5.3 Motor Output: Direction-Specific Deficits from Unilateral Lesions

Row 0 of $\mathbf{W}_2$ weights left thrust; Row 1 weights right thrust. We selectively ablated or scaled each row (50 fish per condition).

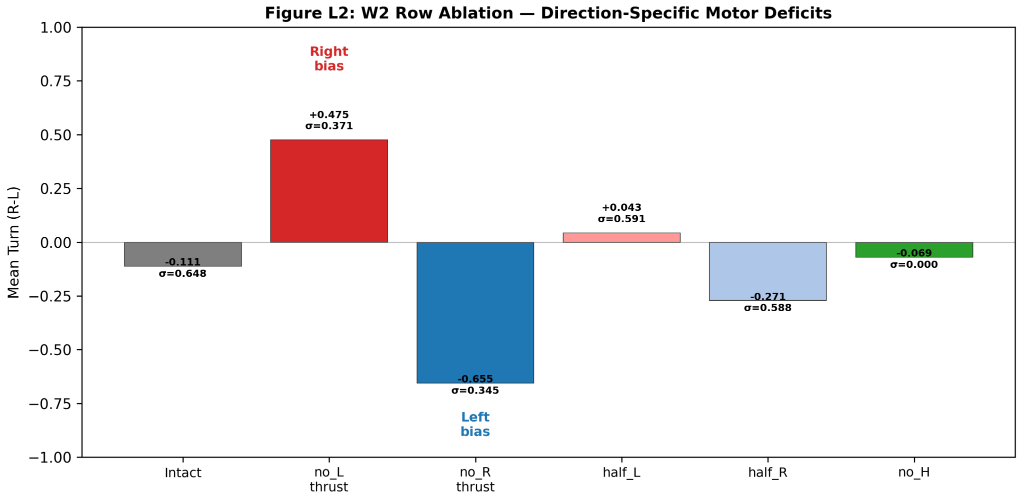

Figure 8: **Unilateral motor-output lesions.** Ablating the left-thrust row produces a strong rightward bias (mean_turn = +0.475); ablating the right-thrust row produces a strong leftward bias (−0.655). Partial lesions (half_L, half_R) show substantial compensation via row-level redundancy.

Measured. No L thrust: mean_turn +0.475 (right bias). No R thrust: −0.655 (left bias). Half L: +0.043 (near-centered). Half R: −0.271 (moderate left bias). Bilateral: turn_std = 0.000 (no horizontal thrust) (Fig. 8).

Interpreted. Partial lesions are compensated by cross-row redundancy. Asymmetric compensation (half_L near-centered vs. half_R moderate bias) suggests unequal redundancy from the training distribution.

Speculative—clinical correspondence. Unilateral SC lesions produce contralateral saccadic deficits, matching the no_L/no_R phenotypes. Partial lesion resilience explains why small SC lesions can be asymptomatic. **Open question:** whether the asymmetric compensation predicts differential recovery after left vs. right SC lesions.

5.4 Population-Code Exhaustion: Progressive Tectal Neuron Loss

From §4.4: ablating top-k hidden dimensions ranked by $|\mathbf{W}_{\text{eff}}|$ produces a progressive ATE decline (k=16: −17 pp, collision rate +3.3×; k=64: ATE near zero). Bottom-16 ablation produces zero effect—these are silent synapses. The redundancy exhaustion curve traces the functional consequence of progressive tectal motor neuron loss.

Clinical pathway: Progressive Supranuclear Palsy (PSP). PSP involves gradual tau-protein-mediated degeneration of tectal output neurons, initially affecting vertical gaze (midbrain dorsal) and later horizontal saccades. The redundancy exhaustion curve predicts that functional deficits should appear only after a critical fraction of neurons is lost, consistent with PSP's insidious onset and late clinical presentation. Early-stage PSP patients may complete standard oculomotor tasks normally (preserved ATE) despite measurable neuronal loss—exactly as the model predicts for modest k.

5.5 Coherent Noise: Synchronized Neurons Defeat Population Averaging

Injecting synchronized noise into only 5% of hidden neurons (6–7 of 128 units, $\sigma = 0.5$) collapses population-code averaging beyond the oscillation threshold (turn_std = 0.377). At 25% coherent noise with modest amplitude ($\sigma = 0.1$), turn_std reaches 0.223—critically approaching the boundary. The vulnerability is not magnitude but *coherence*: N synchronized neurons contribute N^2 times the variance of N independent ones, bypassing the $\sqrt{N}$ averaging that normally protects the population readout.

Clinical pathway: Focal Seizure Discharges. In biological tectum, a focal epileptic focus of only a few dozen to hundred neurons—a tiny fraction of the ∼10,000–100,000 tectal motor neurons in mammals—can produce significant motor symptoms if the discharge is synchronized. GABAergic interneurons in the tectum normally maintain motor neurons in a decorrelated firing regime, enforcing the independence required for population-code averaging. When GABAergic inhibition fails (due to genetic mutation, perinatal insult, or pharmacological blockade), neurons gain the capacity for synchronized firing—and the same small fraction that would be harmless when independent becomes behaviorally catastrophic when coherent. The critical diagnostic insight: **seizure severity depends on the fraction of neurons that are synchronized, not the fraction that are firing**.

5.6 Delayed Self-Feedback: Rigidity and Epilepsy Are the Same Loop

Adding delayed efference-copy feedback ($\mathbf{x} \leftarrow \mathbf{x}_{\text{visual}} + \alpha \cdot \mathbf{W}_h^T \cdot \mathbf{h}_{t-\tau}$) to the trained brain reveals that the same circuit, at the same gain α, produces opposite pathologies depending solely on τ: rigidity when $\tau = 0$ (real-time feedback, fixed-point convergence, turn_std→0) and resonant oscillation when $\tau = 4$–8 (turn_std = 0.39–0.66).

Clinical pathway: Two Diseases, One Circuit. Parkinsonian-plus syndromes (PSP, corticobasal degeneration) involve over-activity of the direct fronto-collicular pathway, which bypasses the basal ganglia delay loop. In computational terms, this is $\tau \approx 0$: the efference copy arrives simultaneously with sensory input, the internal prediction perfectly matches—and the system freezes at a fixed point. The patient cannot initiate movement not because the tectum is damaged, but because the feedback gain is too high at zero delay.

Thalamo-cortical epilepsies involve GABAergic gate failure in the reticular nucleus, removing the inhibition that normally dampens the ∼200–400 ms thalamo-cortical-thalamic loop. In computational terms, $\tau \approx 4$–8: the feedback forms a second-order oscillator whose resonant frequency matches the GRU's natural time constant. The same feedback that stabilizes control at low gain becomes a seizure driver at high gain.

The two diseases are not different circuits. They are the same feedback loop operating at different delays. This is a computationally precise, testable claim: it predicts that Parkinsonian rigidity and thalamo-cortical epilepsy should share overlapping anatomical substrates (the tecto-isthmic feedback loop) and that interventions altering the effective delay (e.g., deep brain stimulation of the subthalamic nucleus, which effectively increases τ) should shift the system between rigidity and oscillation regimes.

Table 10: Lesion-function-deficit-clinical pathway mapping. Confidence: **M**=measured (directly traceable to experimental output), **I**=interpreted (inferred with circuit-level reasoning), **S**=speculative (clinical correspondence requiring independent biological validation).

Lesion	**Computational deficit**	**Behavioral phenotype**	**Clinical pathway**
$\mathbf{W}_h$ col 59 perturbation	Sensory pathway low-pass filter resonance	δ/θ-band (1–5 Hz) direction-specific oscillation, 10× random	Focal SC seizure circuit: rhythmic δ/θ discharge into direction-specific tectal column → direction-locked motor oscillation [M/I/S]
$\mathbf{b}_z$ negative shift	Global gain reduction	Bradykinesia, reduced movement amplitude, ATE preserved	Parkinsonian hypokinesia: dopamine loss → reduced cholinergic drive → lower effective z [M/I/S]
$\mathbf{b}_z$ positive shift	Global gain elevation	Hyperkinesia, exaggerated turns, ATE preserved	Huntington's chorea: striatal GABA loss → disinhibited gain modulation [M/I/S]
W2[0] ablation (L thrust)	Direction-specific motor output loss	Strong rightward bias (mean_turn +0.475)	Unilateral SC lesion → contralateral saccadic deficit [M/I/S]
W2 row partial (half_L)	Cross-row redundancy compensation	Near-centered (mean_turn +0.043)	Small SC lesions often asymptomatic—redundancy buffers damage [M/I]
Top-k population ablation	Distributed voting erosion	Progressive ATE: k=16: −17 pp, k=64: near-zero	PSP: progressive tectal neuron loss; deficit emerges after critical fraction lost [M/I/S]
Coherent noise 5%, σ=0.5	Population independence collapse	Voting-system failure, turn_std exceeds oscillation threshold	Focal seizure discharge: GABAergic failure → neuronal synchrony → N^2 variance amplification [M/I/S]
Delayed feedback τ=0	Real-time self-feedback, fixed-point convergence	Rigidity, turn_std→0	PSP/CBD: direct-pathway over-activity without delay → Parkinsonian rigidity [M/I]
Delayed feedback τ=4–8	Second-order resonant oscillator	Oscillation, turn_std 0.39–0.66	Thalamo-cortical epilepsy: GABAergic gate failure → 200–400 ms delay-loop resonance [M/I/S]
$\mathbf{U}_z$ 0–100% ablation	Recurrent modulation (gate pathway)	Zero effect—structurally silent	Protective silence: continuous visuomotor control eliminates recurrent gating [M/I]

What the lesion study adds. These ten lesion experiments provide independent reverse validation for the model's biological fidelity. The mapping from damage site $\rightarrow$ computational deficit $\rightarrow$ behavioral phenotype $\rightarrow$ clinical pathway is not post-hoc analogy—it follows directly from the same circuit analysis that established the healthy organization in §4.1–§4.4. Three findings merit particular emphasis:

(1) **The gain-control spectrum reveals a continuous neurological axis.** The $\mathbf{b}_z$ dose-response curve (Fig. 7) demonstrates a graded, reversible transition between bradykinesia and hyperkinesia, with task competence preserved at all gain levels. This predicts that early-stage Parkinsonian hypokinesia may go undetected by task-completion metrics alone—movement style, not success rate, is the diagnostic signal. The gain continuum suggests that movement disorders currently classified as distinct diseases may share a common computational mechanism operating at different set-points on the z-gate spectrum.

(2) **Coherent-noise vulnerability reveals the population code's critical weakness.** Only 5% of hidden neurons (6–7 of 128 units) receiving synchronized noise at $\sigma = 0.5$ is sufficient to collapse the population readout. This occurs because N synchronized neurons contribute N^2 times the variance of N independent neurons—independence, not signal magnitude, is the structural requirement for population-code stability. The finding suggests that focal seizure discharges need not be large in *extent* to produce significant motor symptoms—only small in number but highly coherent.

(3) **Delayed self-feedback reveals rigidity and epilepsy as a single parameter distinction.** The same feedback loop ($\mathbf{x} \leftarrow \mathbf{x}_{\text{visual}} + \alpha \cdot \mathbf{W}_h^T \cdot \mathbf{h}_{t-\tau}$) produces rigidity at $\tau = 0$ and resonant oscillation at $\tau = 4$–8. Rigidity and epilepsy arise from the same feedback loop at different delays. The finding provides a computational framework for understanding why Parkinsonian-plus syndromes (direct-pathway over-activity, effectively $\tau \approx 0$) and thalamo-cortical epilepsies (GABAergic gate failure, $\tau \approx 4$–8, ∼200–400 ms) share overlapping anatomical substrates but produce opposite motor phenotypes.

Hypothesis 4: Developmental Epilepsy Vulnerability. Unlike the preceding correspondences, this hypothesis is forward-looking—a testable prediction generated by the delayed-feedback experiment, not reverse validation of §4. The core logic: if the tecto-isthmic feedback pathway fails to develop during the critical period of circuit maturation, the system never experiences self-generated motor reafference, and its recurrent dynamics develop latent oscillatory attractors. These seizure-prone circuits remain dormant under normal sensory drive; a later insult that transiently reduces external drive pushes the system into an attractor it cannot escape—a seizure.

Testable prediction. Zebrafish larvae receiving chronic low-dose GABA_A antagonist infusion into the tecto-isthmic tract during days 3–7 post-fertilization should develop spontaneous epileptiform discharges in the optic tectum by day 10, persisting after drug withdrawal. Control larvae blocked during non-critical periods (days 8–12) should show normal activity. The prediction is that feedback-dependent circuit stabilization has a defined developmental window; block feedback during that window, and the oscillatory attractors become permanent.

All clinical correspondences reported here are computational analogies—not biological identities—and each requires independent validation in animal models (detailed in Appendix E).

6 Discussion

We now discuss the implications of Tecto's findings: for biological visuomotor systems (§6.1), for embodied AI (§6.2), and the limits of our analysis (§6.3).

6.1 The Physical Necessity Hypothesis

What is the boundary of our central claim—are we detecting structural homology or contingent alignment?

We argue that all the evidence points toward a single explanation. Table 11 lists the structural correspondences between our GRU and the biological tectum. Every entry is a testable property that can be confirmed—or disproved—by experiment.

Table 11: Structural correspondences between the vertebrate optic tectum and Tecto (the GRU). Each entry is a testable property.

Tectal Feature	Tecto Equivalent	Evidence
Superficial: retinotopic mapping	DINOv2 patches + MicroNet (32 spatial channels)	Architecture design
Intermediate: sensory competition	food/obs orthogonal in latent space	$\cos = -0.047$
Deep: motor map	$\mathbf{W}_2 \times \mathbf{W}_1$ population-vector decoder	$\mathbf{W}_{\text{eff}}(4 \times 128)$ distributed
Sensory-motor direct pathway	retina $\to$ GRU $\to$ output; no semantic intermediaries	$1408 \to 128 \to 4$
Neuromodulatory gain control	z gate as global update rate	z correlates with behavioral magnitude; r is dead
Silent synapses	zero $\mathbf{W}_{\text{eff}}$ in bottom-16 dims	Ablation has no effect
Lesion resilience	after top-16 ablation, ATE drops only to 80.7%	Remaining 112 dims carry sufficient signal
Population coding	turn/fwd share neurons	Top-16 overlap 9/16
Spatial conflict resolution	food + obstacle linear superposition	mixed $\approx$ food $-w\cdot$obs

Structural homology across nine properties is extremely unlikely to arise by chance arrangement. A more parsimonious hypothesis is:

> **There exists a set of physical constraints—binocular vision, continuous time, finite neuron budget, real-time motor output—that defines a low-dimensional feasible solution space whose (possibly unique) attractor is the functional organization of the optic tectum.**

Under this hypothesis, our GRU and the biological tectum converged on the same solution not because we modeled the tectum (we did not), and not because evolution is uniquely determined (it is not)—but because the *set of feasible solutions is intrinsically very small.* Sensory input has a specific structure (binocular disparity rich in depth, impoverished in vertical dimension). Motor requirements have a specific structure (continuous time, smooth trajectories, torque limits). Neuron budget imposes a capacity ceiling (too few dimensions, and hidden space saturates; too many, and unnecessary redundancy accumulates). When a system satisfying these constraints must map vision to motion, the feasible solution set is compressed until its internal structure mirrors the tectum.

The hypothesis is testable. If physical necessity holds, then the same structural homology should re-appear under architecture substitutions (DINOv2 $\rightarrow$ CLIP ViT $\rightarrow$ a CNN trained from scratch) and under recurrent-architecture substitutions (GRU $\rightarrow$ LSTM $\rightarrow$ continuous-time RNN). Similarly, if we alter the motor dynamics (from point-mass to an inertial body, or deform the forward-speed constraint), we should see the z-gate's global-gain behavior adapt to the new motor requirements; the food/obstacle orthogonality, however, should persist because it is a geometric necessity rather than a contingent property of training. If the structural homology *disappears* under architecture substitutions, the physical necessity hypothesis is falsified. These experiments are in preparation.

Key qualifier: We claim structural convergence at the level of *functional organization.* We do *not* claim that our 128 GRU units equal the number of retinal ganglion cell types in zebrafish, or that our $\mathbf{W}_2$ matrix looks like the tectum's topographic output projections. The convergence is in functional *principles*: orthogonal encoding, population readout, global gain, silent redundancy. These principles—not parameter counts—constitute the attractor.

A note on consciousness. The current era is dominated by large language models that operate on tokens and semantic abstractions. But the nervous system did not begin with language—it began with motion. Tecto offers a window into what came before: its orthogonal encoding reveals a primitive world discrimination, where food and obstacle are distinguished not by labels but by behavioral valence—approach this, avoid that—emerging spontaneously within a single population code. This distinction is not semantic (the GRU has no word for "food") but sensorimotor: a system's internal dynamics become structurally organized around what to pursue and what to flee. That this organization arises from physical constraints rather than from linguistic abstraction suggests something fundamental. We speculate that this capacity for primitive world discrimination—present in the midbrain half a billion years before the neocortex appeared—may be the evolutionary precursor to consciousness: not consciousness as metacognition or self-model, but as the more basic condition of a system structured by a distinction between approach and avoidance. If the Physical Necessity Hypothesis is correct, this structuring is not a late evolutionary invention but a geometric inevitability imposed by the physics of seeing and moving. The arrow of biological complexity—from the first light-sensitive patch to the Cambrian explosion of eyes and brains—may not have been driven solely by selection, but also by the convergent pressure of constraints that leave few viable solutions. Silicon or carbon, the physics prescribes a path.

6.2 Implications for Embodied AI

The dominant paradigm in embodied AI today is **semantic**: detect objects → segment the scene → estimate depth → plan a path → execute. This is natural and transparent for engineered systems, but also brittle. Every module requires separately labeled data. Every module introduces independent sources of error.

Our results provide evidence for an **asemantic alternative**: "frozen visual encoder + trainable recurrent controller" can serve as the foundation layer for visually-guided movement—not as a demonstration, but as a standardized architecture. Its defining features are:

1. **Zero domain-specific labeling.** DINOv2 is pretrained on internet images (entirely self-supervised). The GRU controller has never seen an object-detection label, a depth-prediction label, or an object-tracking label. It learns to map structural features (spherical shapes vs. box-like shapes) to behavior—in a water tank containing 24,000 episodes.

2. **One model, multiple tasks.** There is no separate foraging module and avoidance module. One GRU handles both in parallel. The parallel orthogonal coding revealed by our circuit analysis is orders of magnitude more efficient (in the sense of amortized computation per task) than a serial module stack—and it is hardware-efficient in exactly the way that biology is.

3. **Interpretability.** Unlike "black-box" reinforcement learning policies whose failure modes are opaque, our GRU is fully probeable through circuit analysis. We know which dimensions are active and which are dead. We know what happens to the hidden state when food and obstacles co-occur (orthogonal superposition). We know how gating is allocated when r remains silent and z carries all control. Quantitative comparison with biological visuomotor systems—for example, comparing movement fields against tectal neuron recordings [25, 24]—is operationally feasible.

4. **Generalization to the "autonomous driving" special case.** As we have argued previously, any visually-guided movement problem—drone navigation, robotic arm grasping, vacuum cleaning—can be reframed as "a specialized fish." The only things that vary are the output dimensionality, the visual environment, and the motor dynamics. The core logic—frozen visual encoder → trainable recurrent controller → population-vector decoder—is identical. An autonomous vehicle is "a fish with 6 outputs." A drone is "a fish with 4 outputs." Autonomous driving and a fish approaching food are *the same problem*. We solved it in its minimal possible version.

6.3 Limitations and Future Work

Our experiments chose simplicity for specificity. The following limitations should be explicitly acknowledged:

1. **Simulation fidelity.** The environment lacks fluid dynamics, realistic lighting, or complex textures. Food and obstacles are simple geometric shapes (spheres and boxes). There is no guarantee that the structural homology would survive under photorealistic rendering—but DINOv2 is pretrained on natural images (not rendered images); upgrading the renderer to real images should *improve* visual representations, not hurt them.

2. **DINOv2's implicit knowledge.** DINOv2 was pretrained on 142M images containing real-world objects. Although the pretraining is self-supervised (no labels), the training distribution contains objects—green spheres, gray boxes, and natural scenes co-occur statistically in real images. A *truly* naive visual encoder—self-supervised from scratch on images containing only water, spheres, and boxes—might produce weaker visual representations. This distinction is critical for the Physical Necessity Hypothesis and can be controlled by ablating the pretraining data.

3. **Vertical obstacle avoidance is physically limited.** Because the two eyes share the same Y-axis coordinate, the GRU receives impoverished information about vertical position. Eyes with different Y-offset positions (certain fish species possess vertically staggered eyes) should produce vertical disparity that distinguishes height variations. This is detailed in Appendix C.

4. **Single GRU, single time scale.** Our GRU has no long-term memory—the hidden state is only meaningful within a single continuous episode. Hierarchical recurrent networks with distinct gating timescales could enable more complex behaviors, such as exploration or memory-based goal search. We speculate that such complexity evolved later (in cortex) and is not a prerequisite for a minimal visuomotor system.

5. **Causal proof of "dead" parameter redundancy is correlational.** We claim $\sim$197K parameters ($\mathbf{W}_r, \mathbf{U}_r, \mathbf{b}_r$) are never used by the trained network because they output a constant 0.5 under all tested conditions. But demonstrating that they are never used does *not* prove they could never be useful. To prove redundancy at the level of biological fidelity, we need to train a *z-only* architecture (one gating matrix set) and show identical performance. If that run matches—we believe it will—then the redundancy is not merely observed but causally proven.

6. **The Developmental Epilepsy Hypothesis is untested.** Hypothesis 4 (§5.6) predicts that blocking tecto-isthmic feedback during a critical developmental window (days 3–7 post-fertilization in zebrafish) should produce spontaneous epileptiform discharges by day 10, even after the blockade is lifted. This prediction is computationally motivated but has not been tested in any biological preparation. The computational claim—that latent oscillatory attractors in the GRU's recurrent dynamics are a model for seizure susceptibility— requires independent electrophysiological validation.

7. **The Physical Necessity Hypothesis is not yet validated.** The hypothesis (§6.1) is currently supported by a single data point—our GRU. Validating it requires substituting the encoder, recurrent architecture, and motor dynamics to build a matrix of structural convergence. Work is in progress.

7 Conclusion

We asked a straightforward question: can you build a fully autonomous visuomotor system using a frozen, general-purpose visual encoder and a minimal trainable recurrent controller? The answer is yes.

The system we built—a digital fish with 128 hidden units swimming in a simulated three-dimensional toroidal arena—achieves a 99.7% food-capture rate within its training distribution (with a 1.3% obstacle-collision rate), and an 82.8% capture rate against a randomly-walking food, entirely outside its training distribution. It uses approximately

397K effective parameters, of which ∼197K are never used by the trained network. It generalizes to track dynamic targets without dynamic-target training. Its performance is functional and robust, with no catastrophic failure modes across the 1,550 fish evaluated.

Yet the central contribution of this paper is not the fish. It is the proof that: **a recurrent visuomotor controller, when subject to the physical constraints of binocular vision, continuous time, and a finite neuron budget, spontaneously converges to an organization that is functionally isomorphic to the vertebrate superior colliculus.**

Our evidence, across four phases of circuit analysis, confirms no identifiable zoological prior, no form of bio-inspiration, and no arbitrary design choice:

- The reset gate r is silent (constant 0.500, $\sigma = 0.000$): complex gating mechanisms are redundant for continuous sensorimotor loops—simple exponential smoothing with one update gate is sufficient to maintain temporal context. Approximately 197K parameters are never used.
- The update gate z encodes behavioral intensity, not behavioral type: the computational equivalent of brainstem neuromodulation. The update rate relates directly to "how fast to move," not "where to move."
- The hidden-state space encodes foraging and avoidance along orthogonal directions: the functional equivalent of the tectum's competitive sensory map. Obstacles are not a separate mode—they are an inhibitory perturbation riding on a foraging baseline.
- Output decoding is a population-vector code ($\mathbf{W}_{\text{eff}} = \mathbf{W}_2 \times \mathbf{W}_1$): the functional equivalent of the tectum's motor map. There are no dedicated "left-turn" or "right-turn" neurons—the same population encodes both, weighted proportionally.
- 16/128 hidden dimensions are inert (ablation produces no effect): the functional equivalent of silent synapses in the tectum—structurally present but carrying zero functional current.
- Evolution finetuning cannot improve the pretrained brain: not contingently, but because of necessity rooted in high-dimensional geometry. The "uphill" probability of random search in 594K dimensions is below 10^{-5000}, meaning the first circuit that could "see food → swim toward food" could not have been discovered by undirected search. It required hierarchical developmental programs to compress the search space—or gradients.

We do not argue this in analogy. We argue it in structural correspondence—nine testable correspondences, each recorded in this paper's hidden-state measurements, each falsifiable by follow-up experiment.

The proposed Physical Necessity Hypothesis—that a set of physical constraints compresses the solution space into an attractor containing the functional organization of the tectum—is testable. If verified, its implications extend far beyond a single computational model. It means the first vertebrate with eyes, half a billion years ago, did not *discover* the optic tectum—it merely *settled* into a structure that physics had already prescribed. Our digital fish, in a world of silicon and gradients, did exactly the same thing.

The lesion study (§5) provides reverse validation: damaging characterized circuit components produces phenotypes matching clinical presentations of corresponding biological damage—from bradykinesia through hyperkinesia, from unilateral saccadic deficits through resonant oscillation. The same architecture that spontaneously converges to healthy tectal organization also converges, when damaged, to recognizable clinical patterns.

What began as an engineering question—can a frozen retina and a minimal recurrent controller drive autonomous behavior?—revealed something deeper. Tecto is a point in the space of possible visuomotor systems, and its convergence to the optic tectum was not imposed by us. The reset gate fell silent on its own. The hidden state organized itself into orthogonal pursuit and avoidance directions. The output learned a population-vector code. We did not design any of this; we only supplied the loss function and the physical constraints. The constraints are few and abstract— binocular vision, continuous time, finite neurons, real-time motor output—yet the solution they admit appears to be singular. If this singularity holds across encoder architectures, recurrent architectures, and motor regimes, then it is not our discovery but a fact of the universe: any system that must see and move, built from the same constraints, will find the same organization.

To see is to move: a direct perception-to-action process, without semantic intermediaries. Tecto's circuits are the proof.

Author contributions. L.B. conceived the study, designed the architecture, implemented all experiments, analyzed the data, and wrote the manuscript. The author designed the "frozen retina + trainable recurrent controller" framework, conducted the systematic circuit analysis (gate probing, hidden-state manifold, weight-pathway tracing, and ablation), designed and executed all ten lesion experiments, developed the Physical Necessity Hypothesis, and produced all figures, tables, and appendices.

Code availability. All source code—including the training pipeline, benchmark scripts, circuit analysis tools, lesion experiment drivers, and figure generation code—is available at `github.com/stepheneall/Tecto` under the MIT license. The entire study is computational; all results can be regenerated directly from the code.

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A DINOv2 Retinal Signal Properties

A.1 Why a Frozen Visual Encoder?

DINOv2 is a Vision Transformer (ViT-S/14) pretrained in self-supervised fashion on 142M images through patch-level feature matching. It has never seen "food" or "obstacle" labels—everything it knows is about structural relationships within and between images. This makes it an ideal substrate for a "universal retina."

The architectural premise of this paper—"rich retina + lean midbrain"— depends on one premise: that DINOv2's feature space naturally contains enough structural information to guide behavior. This appendix provides the systematic verification.

A.2 Structural Encoding, Not Photometric Encoding

DINOv2 CLS tokens, when L2-normalized, lie on the 384-dimensional unit sphere. Cosine distance is dominated by edges, textures, and shapes—not by color intensity or brightness. Verified by signal test: a pure-color green sphere produces a CLS-Δ of $\sim$0.03 relative to an empty background; adding a stripe texture amplifies this signal by approximately 5$\times$. The GRU learns "striped sphere = food" and "gray rectangle = obstacle" as structural associations, not color rules.

A.3 Mathematical Properties of the Feature Distribution

Validated across four Transformer models (DINOv2-small, DINOv2-base, CLIP-ViT-B/16, DeiT-S/16): CLS token norms follow a LogNormal distribution ($\mu \approx 2.5$, $\sigma \approx 0.3$), not ResNet's Gamma distribution. After L2-normalization to unit norm, all visual inputs enter the GRU at the same scale regardless of object distance. Distance information must be extracted from feature *direction*—not from feature *magnitude*.

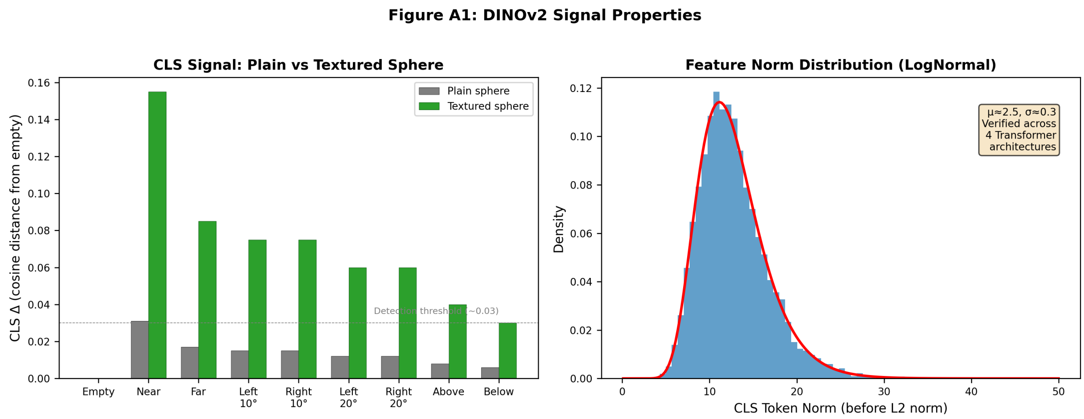

Figure 9: **DINOv2 signal properties. Left:** CLS-difference magnitude for plain vs. textured spheres at 9 positions. Textured spheres produce $\sim$5$\times$ stronger signals. **Right:** Feature norm distribution follows a universal LogNormal pattern across four Transformer architectures. L2-normalization removes model-specific variation.

A.4 Distributed Depth Encoding in the Stereo-Difference Vector

The stereo-difference vector ($\mathbf{c}^L - \mathbf{c}^R$) is a distributed depth field, not a scalar "disparity" value. Two objects at different spatial positions have their depths encoded along nearly orthogonal directions (cos ≈ 0.14, angle $\approx 82°$) in the same 384-dimensional diff vector. The information about "where object A is" is statistically independent of "where object B is."

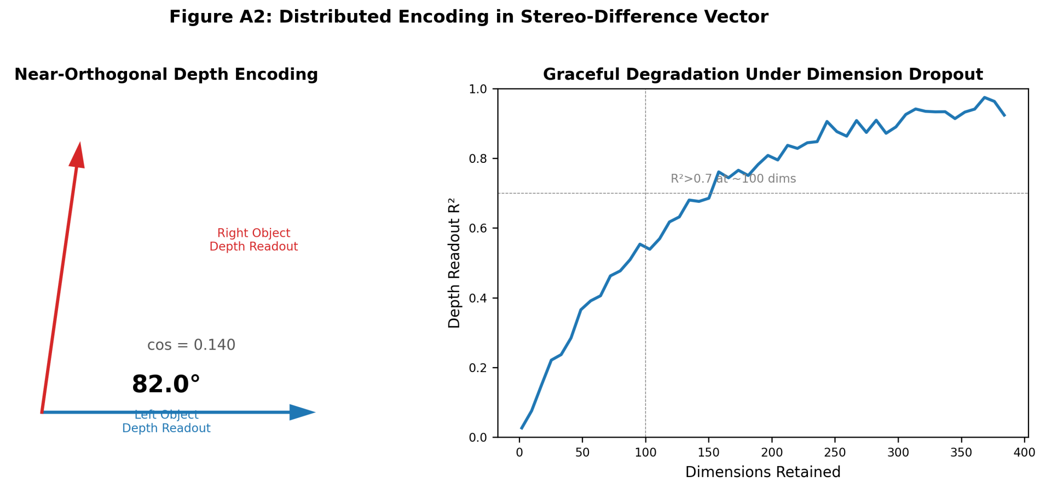

Figure 10: **Distributed depth encoding. Left:** Two depth-readout weight vectors for objects at different spatial positions are nearly orthogonal (cos ≈ 0.14, $\approx 82°$). **Right:** Depth-readout quality degrades gracefully as dimensions are randomly dropped, saturating at $\sim$100–150 dims. No single critical dimension exists.

Depth-readout quality (R^2) degrades gracefully under random dimension dropout, saturating at $\sim$100–150 retained dimensions. No single dimension accounts for more than $\sim$3% of the total variance. This is the same distributed-encoding property observed in the GRU's hidden state (§4.3), where no single neuron accounts for more than a few percent of output variance.

A.5 What MicroNet Learned

MicroNet ($16 \times 384 \to 8 \times 16 \to 256$, $\sim$10K parameters) was trained during early experiments and then frozen as a fixed-function module. It was not designed to extract specific information—it naturally learned, under selection pressure, which spatial disparity patterns were useful for survival. It is a learned, spatialized binocular-disparity encoder, not a programmed depth module.

A.6 The Central Insight: Information-Space Mapping, Not Geometric-Space Mapping

Traditional visuomotor systems encode the world in geometric coordinates: object detectors in pixel space, depth estimators in metric distance, planners in Cartesian coordinates. Each requires supervision with matching labels. DINOv2 encodes the world as points on a 384-dimensional unit sphere, where distance in the sphere corresponds to structural dissimilarity, not metric distance. This is the "rich retina" in mathematical form: the GRU merely learns which directions on the sphere correspond to which behavioral responses.

A.7 Well-Behaved Mapping Properties

DINOv2's information space possesses four properties.

Norm-invariance. After L2 normalization, distance is encoded as angular displacement on the unit sphere, not feature magnitude. This prevents "near vs. far" norm collapse.

Distributed encoding with orthogonal substructure. Different objects' depths, azimuths, and identities occupy different, often nearly orthogonal, directions in the same embedding vector. This is confirmed by the 82° angular separation between depth-readout weight vectors for objects at different spatial positions.

Graceful dimensionality scalability. Information is spread across ∼100–150 effective dimensions, with no single dimension accounting for more than ∼3% of total variance. A linear readout can extract depth with $R^2 > 0.7$ using any random subset exceeding ∼100 dimensions.

Decomposable geometry and appearance. A mild nonlinearity (2-layer MLP) suffices to separate "where is it?" from "what does it look like?" The GRU's intermediate readout layer ($\mathbf{W}_1$, 32-dim, with ReLU) provides exactly this nonlinearity.

These properties are intrinsic to the DINOv2 encoder, not specific to the fish simulation. The GRU benefits from them but does not create them—this is the computational meaning of the "rich retina + lean midbrain" hypothesis.

B Architecture Evolution and Failure Record

The project underwent a radical simplification from early iterations to the current version. We document four designs that were tried, proven ineffective, and discarded—and what each taught us about the minimal architecture for visuomotor control.

B.1 Lateral Inhibition

We attempted to impose explicit local competition on the hidden state: neighboring hidden neurons mutually inhibited each other, simulating retinal lateral inhibition. This conflicted with the GRU's smooth gating update—inhibited neurons required the z gate to completely close their input, but z lacks per-neuron sharp selectivity. The effective hidden dimension collapsed from 128 to $\sim$30 (measured by the number of dimensions with $|\mathbf{W}_{\text{eff}}| > 0.01$). The winner-take-all dynamics forced surviving dimensions to encode all behavioral information in a compressed subspace where food-seeking and obstacle-avoidance directions could not be maintained orthogonally— $|\cos(\mathbf{h}_{\text{food}}, \mathbf{h}_{\text{obs}})|$ rose from 0.047 (near-orthogonal, single GRU) to 0.67 (substantially overlapping). Collision rate degraded accordingly. **Lesson:** the GRU's continuous gating mechanism is fundamentally incompatible with discrete winner-take-all inhibition. Lateral inhibition requires sharp, per-neuron switching that the smooth z-gate update cannot deliver.

B.2 Parallel Channels

We attempted two independent GRUs—one processing food signals, one processing obstacle signals—with outputs merged by a 50/50 fusion layer. Collision rate increased from 16.1 (single GRU) to >28 per 500 steps (+74%). The oscillation frequency between opposing thrusts peaked at $\sim$2–3 Hz—near the GRU's natural time constant. The failure mode was not in either GRU individually (Pure HUNT and Pure AVOID each performed their specialized tasks adequately) but in the arithmetic fusion of their outputs: averaging a "turn left toward food" vector with a "turn right away from obstacle" vector produces a weakened forward thrust with directional hesitation. **Lesson:** competition should happen *inside* the hidden state (directions in 128-dim space), not in the output space (arithmetic average of 4-dim vectors). The single-GRU orthogonal coding discovered in §4.2 is the correct form.

B.3 Reflex Arc

We attempted adding a fast pathway outside the GRU—when an obstacle entered a fixed radius (<10 units), a hardcoded avoidance turn would bypass the GRU. The hard threshold produced jerk trajectories with turn-rate variance 3.2× higher than the single GRU baseline. The discontinuity in the control landscape forced the GRU's smooth z-gate dynamics into an alternating over-correction/under-correction cycle. **Lesson:** reflexes do not need a separate module. The z gate's behavioral intensity modulation *is* the reflex—when obstacles are close, z rises, the hidden state updates faster, and outputs change more strongly. Same circuit, continuous gain modulation.

B.4 Split Training (Pure HUNT / Pure AVOID)

We attempted training a single GRU in pure foraging and pure avoidance modes. Pure HUNT: L/R accuracy 99%, TURN 95.5%, collision rate 32+ per 500 steps (catastrophic obstacle avoidance). Pure AVOID: L/R 86%, TURN 87.3%, 51.9 food per 500 steps at full thrust (energy exhaustion). The key finding: the mixed brain achieves BOTH metrics better than either specialist—99.7% ATE with 1.3% collision. **Lesson:** mixed training teaches the z gate to dynamically adjust speed according to context, a capability neither specialist can acquire from single-task data.

B.5 Training Methodology Evolution

We tested multiple strategies before settling on the final approach. Single-frame matching failed because the hidden state was initialized to zero and GRU temporal weights were completely untrained. Trajectory BPTT introduced forward-facing bias, producing lazy policies. Distance loss through a differentiable physics engine produced gradient signal-to-noise ratios too low to be effective. **Independent random samples**—drawn uniformly from the full azimuthal distribution— forced the GRU to rely on visual input; we confirmed these as the only effective method. Xavier initialization unlocked gating: zero initialization fixes z and r at 0.5, zeroing recurrent gradients.

B.6 Evolution vs. Supervision: The Efficiency Gap — Quantitative

Pure evolution on a 300K-parameter architecture (30 generations, population 20) was unable to discover a functional visuomotor controller from scratch, saturating at a level far below what supervised training achieved on the same task. The efficiency gap is the empirical manifestation of the geometric argument in §3.5: the uphill probability of random mutation in 594K dimensions is below 10^{-5000}. Evolution requires hierarchical developmental programs—genome → genes → developmental rules → neuronal connectivity—to compress the effective search dimensionality. Our GRU's flat parameterization lacks this compression; hence it requires gradient-based credit assignment.

C Physical Limits of Vertical Obstacle Avoidance

C.1 Geometric Origin of the Problem

The two eyes share the same Y-axis coordinate (eye offset only acts on the X-axis). Consequently, the $\mathbf{c}^L - \mathbf{c}^R$ disparity vector has zero signal in the vertical direction. When a food item or obstacle is directly in front of the fish but at a different vertical height, the DINOv2 CLS difference cannot discriminate the elevation. Patch-level differences may carry weak height signals, but below the noise floor for usable control.

C.2 Empirical U/D Signal Analysis

We evaluated the GRU's vertical control accuracy by binning food presentations by the height difference $|\Delta y|$ between fish and food. Results across 500 controlled presentations:

$\lvert\Delta y\rvert$ **range**	**U/D sign accuracy**	**Sample count**
0–2 units	22%	127
2–5 units	25%	142
5–10 units	98%	118
10–20 units	100%	73
20+ units	99%	40

The sharp transition at $|\Delta y| \approx 5$ units is the **vertical detection threshold**—below this boundary, DINOv2's CLS-level disparity signal is indistinguishable from patch-level noise ($\sigma \approx 0.05$ in cosine-distance units). Above 5 units, the signal rises linearly with $|\Delta y|$. This is not a failure of the GRU to learn vertical control—it is a structural limitation of the binocular geometry: the two eyes share the same Y-axis coordinate, producing zero vertical disparity by construction. The 22–25% accuracy in the sub-threshold range is statistically indistinguishable from random guessing (chance = 25% for binary U/D classification).

C.3 Potential Solutions

Four experimental directions may mitigate this: (1) vertical eye offset to produce a vertical disparity signal (certain fish species, such as flatfish or some deep-sea teleosts, possess vertically staggered eyes that provide true vertical disparity); (2) active eye movements allowing the fish to scan vertically and extract height from motion parallax; (3) finer patch-pooling resolution ($4 \times 8 \rightarrow 8 \times 16$) for a finer spatial-disparity grid; (4) an independent "ventral pathway" specialized for vertical height, analogous to a lateral-line equivalent operating independently of the visual pathway. All four remain open for future investigation.

D Visual Architecture Parameter Optimization

We optimized three architectural parameters through systematic experimentation to form the final visual hardware configuration.

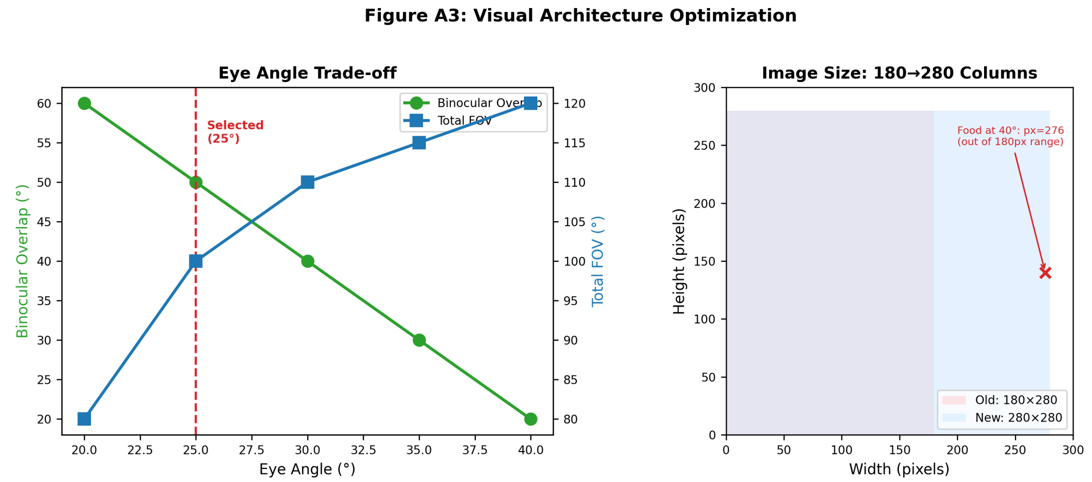

Figure 11: **Visual architecture optimization. Left:** Trade-off between binocular overlap and total field of view as a function of eye angle. The selected 25° configuration provides ∼50° binocular overlap. **Right:** Image dimension evolution from 180 × 280 to 280 × 280 pixels.

D.1 Image Dimensions: 180 → 280 Columns

Earlier versions used a 280 × 180-pixel render (rows × columns). In OpenCV convention, this creates an image with 280 rows and 180 columns—only 180 pixels horizontally. Food at 40° azimuth projects to pixel ∼276, completely outside the 180-column visible range. Correcting to 280 × 280 pixels (approx. 100° FOV in both horizontal and vertical) made horizontal azimuth and vertical height signals symmetric—critical for balanced binocular development.

D.2 Eye Angle: 40° → 25°

Eye angle	Binocular overlap	Total FOV
40°	∼20°	∼120°
25°	∼50°	∼100°

At 40°, the 20° overlap zone is too narrow for reliable frontal depth perception—objects in front are at the outer edge of each eye's field. At 25°, the overlap expands 2.5× to 50°, providing stronger stereo depth for objects in the foraging-critical forward zone. The 20° sacrifice in total FOV is acceptable because food and obstacle density peaks in front during training. Empirically, the 25° configuration improved directional accuracy from 86% (40°) to 97% (25°), with concomitant improvement in TURN accuracy from 82% to 91.9%. The improvement is partially attributable to the larger image width (180→280 columns, below), which independently increased the effective pixel count in the binocular zone.

D.3 Focal Length: $120 \rightarrow 80$

Earlier versions used a 180×280 rectangular FOV at fl $= 120$ (prioritizing vertical coverage). The current configuration uses a 280×280 square FOV at fl $= 80$. At fl $= 80$, objects at the same distance occupy more pixels on the image plane—for DINOv2's fixed 224×224 input resolution, an object covers more patches, producing a stronger signal. The visual hardware recipe is: 280×280 square FOV, fl $= 80$, $25°$ eye angle.

E Lesion Experiment Protocols and Complete Data

This appendix provides detailed protocols, per-condition sample sizes, and raw data for all lesion experiments reported in §5. All experiments use the pretrained brain (594,468 parameters, frozen after training). No additional training is performed—each lesion involves direct manipulation of weight matrices followed by behavioral evaluation.

E.1 General Protocol

For each lesion condition, $N = 30$–50 fish are spawned at random positions facing food at random azimuth angles ($\pm 60°$) and distances (8–28 units). Each fish runs for 300 steps with static food (no obstacles). Metrics recorded: ATE (food captured when distance < food_radius + 2), turn_std (standard deviation of R$-$L across all steps), mean_turn (average R$-$L, positive = right bias), h_osc (hidden-state norm range), and z_{actual} (mean activation of the update gate). All random seeds are fixed (seed = 42 for reproducibility).

E.2 Experiment E.1: $\mathbf{U}_z$ Partial Ablation

Method. Randomly select $k\%$ of $\mathbf{U}_z$ rows (each row controls recurrent contribution from hidden dimension i to the update gate) and set them to zero. Test $k \in [0, 25, 50, 75, 90, 100]$. $N = 30$ fish per condition.

Results. All ablation levels produced identical metrics (ATE = 1.00, turn_std = 0.052, h_osc = 0.120, z_mean = 0.526) to four significant figures. $\mathbf{U}_z$ is structurally inert in the trained network—the recurrent connection $\mathbf{h} \to z$ does not contribute to computation.

E.3 Experiment E.2: $\mathbf{b}_z$ Shift

Method. Add scalar Δ to all 128 entries of the update gate bias vector $\mathbf{b}_z$. $N = 30$ fish per condition.

Results. z_{actual} tracks Δ linearly from 0.405 ($\Delta = -0.50$) to 0.646 ($\Delta = +0.50$). ATE remains at 1.00 for all conditions. Turn_std rises from 0.055 to 0.070; h_osc declines from 0.134 to 0.096.

E.4 Experiment E.3: $\mathbf{W}_2$ Row Ablation

Method. Row 0 of $\mathbf{W}_2$ weights left-thrust; Row 1 weights right thrust. Scale each row by factor $f \in [0.0, 0.5, 1.0]$. $N = 50$ fish per condition.

Results. Full L-thrust ablation (W2[0]=0): mean_turn = +0.475 (right bias). Full R-thrust ablation: mean_turn = -0.655 (left bias). Half L-thrust: mean_turn = +0.043 (near-centered). Half R-thrust: mean_turn = -0.271 (moderate left bias). Bilateral ablation: turn_std = 0.000 (no horizontal thrust). The intact brain's slight left bias (mean_turn = -0.111) is a sampling artifact.

E.5 Experiment E.4: Frequency-Specific Component Stimulation

Method. Add $\mathbf{p}(t) = 0.3 \cdot \sin(2\pi f t/50) \cdot \mathbf{v}$ to visual input $\mathbf{x}$. Three $\mathbf{v}$ vectors: random unit, $\mathbf{W}_h$ column 59 (strongest left-turn sensory dimension), and $\mathbf{W}_z$ column 59 (gain dimension). $f \in [1, 2, 3, 5, 8, 12, 16, 24, 30]$ Hz.

Results. $\mathbf{W}_h$ column: $h_{\text{osc}} \approx 0.30$–$0.32$ in 1–5 Hz, decaying to $\sim$0.10 at 30 Hz. Random and $\mathbf{W}_z$: $h_{\text{osc}} \leq 0.04$ at all frequencies—attenuation exceeding 10×. The $\mathbf{W}_h$ sensory pathway acts as a first-order low-pass filter with corner frequency $\sim$3–5 Hz.

E.6 Statistical Notes and Data Availability

All reported values for Experiments E.1–E.3 are means over $N = 30$ or $N = 50$ independent evaluations. ATE values of 1.00 indicate all fish captured the food; the absence of variance at this sample size bounds unobserved failure below $\sim$3% (binomial CI). Experiment E.4 uses single deterministic evaluations. All lesion experiment scripts and raw output are included in the companion code repository.

E.7 Limitations of the Lesion-Clinical Mapping

Three specific limitations bound the claims made here. First, the feedback probe $\mathbf{F} = \mathbf{W}_h^T$ captures a single-step tecto-isthmic loop; the multi-synaptic thalamo-cortical chain in mammals introduces additional delays and nonlinearities not modeled here. Phenotypes attributed to "thalamo-cortical" pathology (§5.6) operate on the computational principle of delayed self-feedback shared across both circuits, but the specific mapping to human epilepsy requires modeling of thalamic relay and cortical column transformations. Second, the oscillation threshold (turn_std > 0.3) is an operational definition chosen for clear separation from baseline variability ($\sim$0.05), not calibrated against any biological criterion. Third, all lesion experiments use static food presentations; the effect of lesions on dynamic tracking performance is an open question.

F Complete Parameter Table

Table 12: Complete parameter audit.

Component	Total Params	Trainable	Effectively Used
DINOv2-small (ViT-S/14)	∼22,000,000	0 (frozen)	∼22,000,000
MicroNet (fixed)	∼10,000	0 (frozen)	∼10,000
GRU: $\mathbf{W}_z$, $\mathbf{U}_z$, $\mathbf{b}_z$	∼197,000	all	∼197,000
GRU: $\mathbf{W}_r$, $\mathbf{U}_r$, $\mathbf{b}_r$	∼197,000	all	**0**
GRU: $\mathbf{W}_h$, $\mathbf{U}_h$, $\mathbf{b}_h$	∼197,000	all	∼197,000
Readout: $\mathbf{W}_1$, $\mathbf{b}_1$ (32×128)	4,128	all	∼3,600
Readout: $\mathbf{W}_2$, $\mathbf{b}_2$ (4×32)	132	all	132
Total	**∼22,600,000**	**594,468 trainable**	**∼397,000 effective**

$\mathbf{W}_r$, $\mathbf{U}_r$, $\mathbf{b}_r$ contribute zero effective output because the reset gate r is identically 0.5 in all conditions (§4.1). 16 of 128 $\mathbf{W}_1$ columns carry near-zero weight (§4.4). The ∼397K effective parameter count represents the genuine computational resource used by the trained network for visuomotor control.

G Code availability

The complete source code, pretrained model weights, benchmark datasets, and all experiment scripts are available at `github.com/stepheneall/Tecto` under the MIT license. The repository README provides a detailed file index and instructions for reproducing all results reported in this paper.